



Chapter preview

Chemists use the unit 'mole' to describe the quantity of particles in a substance.

- 12.1 The mole and Avogadro constant
- 12.2 The mole and molar mass
- 12.3 Calculating percentage by mass of an element in a compound
- 12.4 Calculating the mass of an element in a compound
- 12.5 Calculating the relative atomic mass of an element
- 12.6 Determining empirical formulae from experimental data
- 12.7 Determining molecular formulae from experimental data
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Counting in Chemistry

It is easy to count 10 coins by hands. But it would be difficult to count 1000 coins and even 100 000 coins by hands. Apart from counting directly by hands or by using a coin-counting machine, is there any effective way to 'count' a large number of coins?



🕒 How can we count a huge number of \$0.5 coins effectively?

One way to do the counting indirectly is by weighing. Similarly, atoms, molecules and ions are so tiny that it is impossible to count them one by one. Chemists count the huge number of particles in a substance also by weighing.

Think about...

- 1 Given that the mass of a \$0.5 coin is 4.92 g, how many coins are there in 246 kg of \$0.5 coins?
- 2 What unit is used in counting particles, like atoms, molecules and ions, in Chemistry?
- 3 Can we predict the mass of the product formed from a chemical reaction if the masses of the reactants are known?

After studying this chapter, you should be able to answer the above questions.

12.1 The mole and Avogadro constant

In daily life, some **special units** are used to describe the quantity of items. For example, eggs are often packed in dozens and papers are packed in reams. A dozen of eggs refers to 12 pieces and a ream of paper refers to 500 sheets. See Figure 12.1.



Figure 12.1 Eggs are often packed in dozens and papers are often packed in reams.



History corner

Avogadro constant

The Avogadro constant is named after an Italian scientist, Amedeo Avogadro. He proposed that equal volumes of all gases under the same conditions contain the same number of molecules.



Amedeo Avogadro
(1776–1856)

Unlike eggs and papers, particles (i.e. atoms, ions and molecules) are too small to be seen. It is **impossible to count these particles one by one**. Chemists use a special unit called **mole** (abbreviation: **mol**) to describe the quantity of particles in a substance.

Chemists have chosen the number of atoms in exactly 12.0 g of carbon-12 as the **reference unit** for the mole. **The number of atoms in exactly 12.0 g of carbon-12 is 6.02×10^{23}** . This number is called the **Avogadro constant** (symbol: **L**; unit: **mol⁻¹**). As a pure substance can be represented by a formula, its simplest unit is called a formula unit. Hence, we can also say that, one mole of any substance contains 6.02×10^{23} formula units of that substance. See Table 12.1.

Substance	Formula unit
Copper	Cu
Carbon	C
Water	H ₂ O
Sodium chloride	NaCl

Table 12.1 Formula units of some substances

Figure 12.2 gives us ideas on how much one mole of some substances are.



History corner

Mole

The term 'mole' was introduced by a German scientist Wilhelm Ostwald in 1893. It came from the word 'Molekül' meaning molecule. He was one of the winners of the Nobel Prize for Chemistry in 1909.



Wilhelm Ostwald
(1853–1932)

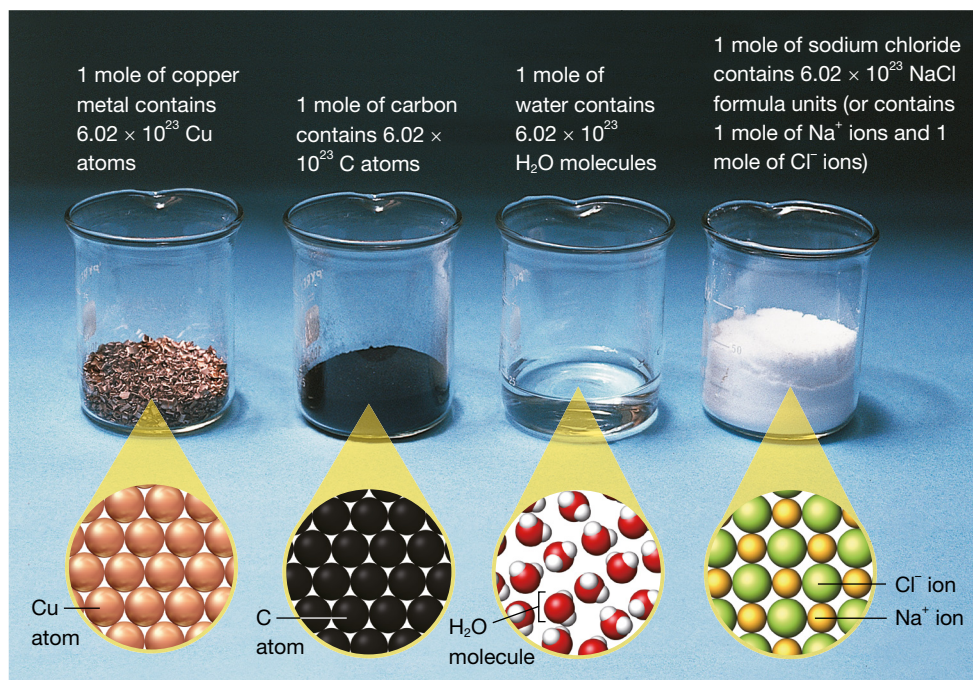


Figure 12.2 One mole of each of the four substances (copper, carbon, water and sodium chloride). They all contain 6.02×10^{23} formula units.

Chemists define mole in the following way:



Key point

One mole is the amount of a substance that contains the same number of formula units as the number of atoms in exactly 12.0 g of carbon-12.



Example 12.1

Calculating the number of moles from the number of formula units and vice versa

- (a) A sample of nitrogen contains 1.024×10^{24} molecules. Calculate the number of moles of nitrogen molecules in the sample.
- (b) Calculate the number of atoms in 3 mol of nitrogen.
(Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

Solution

$$\begin{aligned}
 \text{(a) Number of moles of N}_2 \text{ molecules} &= \frac{\text{number of N}_2 \text{ molecules}}{\text{Avogadro constant (mol}^{-1}\text{)}} \\
 &= \frac{1.024 \times 10^{24}}{6.02 \times 10^{23} \text{ mol}^{-1}} \\
 &= 1.70 \text{ mol}
 \end{aligned}$$

Cont'd

(b) Number of N_2 molecules

= number of moles of N_2 molecules (mol) $\times$ Avogadro constant (mol^{-1})

$$= 3 \text{ mol} \times 6.02 \times 10^{23} \text{ mol}^{-1}$$

$$= 1.806 \times 10^{24}$$

Since there are two N atoms in one N_2 molecule, number of N atoms

$$= 2 \times 1.806 \times 10^{24}$$

$$= 3.612 \times 10^{24}$$

Self-test 12.1

(a) Calculate the number of formula units in 1.5 mol of potassium chloride.

(b) Calculate the number of ions in 1.5 mol of potassium chloride.

(Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)



Class practice 12.1

- A sample of oxygen gas contains 0.5 mol of oxygen molecules.
 - Calculate the number of oxygen molecules in the sample.
 - Hence, calculate the number of oxygen atoms in the sample.
(Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)
- Given that a sample contains 2.408×10^{24} sodium atoms, calculate the number of moles of sodium atoms present.
(Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

12.2 The mole and molar mass



Learning tip

The relative atomic masses of elements can be found from the Periodic Table on the inside front cover of this book.



Think about it

Which has a greater mass, 1 mole of iron atoms or 1 mole of gold atoms?

In Chapter 5, we have learnt that the relative atomic mass of carbon-12 is 12.0. Scientists found that one mole of ^{12}C has a mass of 12.0 g. The mass of one mole of a substance is called its **molar mass**. The unit of molar mass is **gram per mole (g mol^{-1})**.

- For substances consisting of atoms:
molar mass = relative atomic mass expressed in g mol^{-1}

Substance	Symbol	Relative atomic mass	Molar mass
Carbon	C	12.0	12.0 g mol^{-1}
Iron	Fe	55.8	55.8 g mol^{-1}

- For substances consisting of molecules:

molar mass = relative molecular mass expressed in g mol^{-1}

Substance	Formula	Relative molecular mass	Molar mass
Oxygen	O_2	$16.0 \times 2 = 32.0$	32.0 g mol^{-1}
Water	H_2O	$1.0 \times 2 + 16.0 = 18.0$	18.0 g mol^{-1}

- For substances consisting of ions:

molar mass = formula mass expressed in g mol^{-1}

Substance	Formula	Formula mass	Molar mass
Calcium chloride	CaCl_2	$40.1 + 35.5 \times 2 = 111.1$	111.1 g mol^{-1}



Learning tip

Relative atomic mass, relative molecular mass and formula mass carry no units.

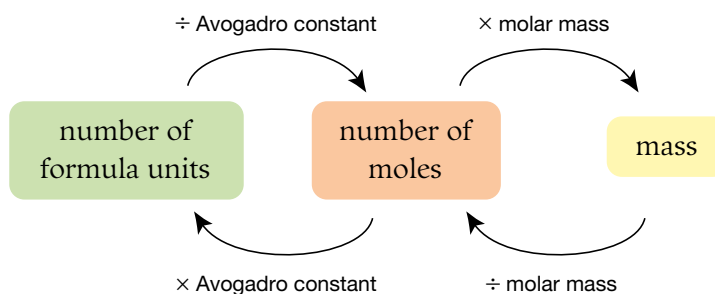


Class practice 12.2

- Calculate the molar mass of each of the following substances.
 - Silver
 - Fluorine
 - Ammonia
 - Ethanol ($\text{C}_2\text{H}_5\text{OH}$)
 - Iron(III) sulphate
 (Relative atomic masses: $\text{H} = 1.0$, $\text{C} = 12.0$, $\text{N} = 14.0$, $\text{O} = 16.0$, $\text{F} = 19.0$, $\text{S} = 32.1$, $\text{Fe} = 55.8$, $\text{Ag} = 107.9$)
- Calculate the mass of each of the following substances.
 - 1 mol of sodium sulphate
 - 0.5 mol of tetrachloromethane (CCl_4)
 (Relative atomic masses: $\text{C} = 12.0$, $\text{O} = 16.0$, $\text{Na} = 23.0$, $\text{S} = 32.1$, $\text{Cl} = 35.5$)

Relationship between number of formula units, number of moles and mass

The diagram below summarizes the relationship between 'number of formula units', 'number of moles' and 'mass' of a substance.



We can calculate one item from another by using the following formulae:

Key point

$$\text{Number of moles (mol)} = \frac{\text{mass (g)}}{\text{molar mass (g mol}^{-1}\text{)}}$$

$$\text{Number of moles (mol)} = \frac{\text{number of formula units}}{\text{Avogadro constant (mol}^{-1}\text{)}}$$



Example 12.2

Calculating the mass and number of formula units from the number of moles

A vessel contains 1.85 mol of methane (CH_4).

- Calculate the mass of methane in the vessel.
- Calculate the number of methane molecules in the vessel.

(Relative atomic masses: $\text{H} = 1.0$, $\text{C} = 12.0$; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

Solution

- Molar mass of $\text{CH}_4 = (12.0 + 1.0 \times 4) \text{ g mol}^{-1} = 16.0 \text{ g mol}^{-1}$

Mass of $\text{CH}_4 \text{ (g)} = \text{number of moles of } \text{CH}_4 \text{ (mol)} \times \text{molar mass of } \text{CH}_4 \text{ (g mol}^{-1}\text{)}$

$= 1.85 \text{ mol} \times 16.0 \text{ g mol}^{-1}$

$= 29.6 \text{ g}$
- Number of CH_4 molecules

$= \text{number of moles of } \text{CH}_4 \text{ (mol)} \times \text{Avogadro constant (mol}^{-1}\text{)}$

$= 1.85 \text{ mol} \times 6.02 \times 10^{23} \text{ mol}^{-1}$

$= 1.11 \times 10^{24}$

Self-test 12.2

Consider a 0.175 mol sample of magnesium hydroxide.

- Calculate the mass of magnesium hydroxide in the sample.
- Calculate the number of hydroxide ions in the sample.

(Relative atomic masses: $\text{H} = 1.0$, $\text{O} = 16.0$, $\text{Mg} = 24.3$; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)



Chemistry in daily life

Mole day

Mole day was proposed by a chemistry teacher in the United States in 1991. It is an unofficial holiday celebrated among chemists or anyone who is interested in Chemistry on every year of October 23, from 6:02 a.m. to 6:02 p.m. The date and time of this 'holiday' were derived from the Avogadro constant (i.e. 6.02×10^{23}).



Let us mark this day down!



Example 12.3

Calculating the mass of a particle of a substance

Calculate the mass of

- (a) 1 Na atom. (b) 1 H₂O molecule. (c) 1 formula unit of NaCl.

(Relative atomic masses: H = 1.0, O = 16.0, Na = 23.0, Cl = 35.5; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

Solution

$$\text{Mass of 1 formula unit} = \frac{\text{molar mass (g mol}^{-1}\text{)}}{\text{Avogadro constant (mol}^{-1}\text{)}}$$

$$(a) \text{ Mass of 1 Na atom} = \frac{23.0 \text{ g mol}^{-1}}{6.02 \times 10^{23} \text{ mol}^{-1}} = 3.82 \times 10^{-23} \text{ g}$$

$$(b) \text{ Mass of 1 H}_2\text{O molecule} = \frac{(1.0 \times 2 + 16.0) \text{ g mol}^{-1}}{6.02 \times 10^{23} \text{ mol}^{-1}} = 2.99 \times 10^{-23} \text{ g}$$

$$(c) \text{ Mass of 1 formula unit of NaCl} = \frac{(23.0 + 35.5) \text{ g mol}^{-1}}{6.02 \times 10^{23} \text{ mol}^{-1}} = 9.72 \times 10^{-23} \text{ g}$$

Self-test 12.3

Calculate the mass of

- (a) 1 Mg atom. (b) 1 Br₂ molecule. (c) 1 formula unit of CaCO₃.

(Relative atomic masses: C = 12.0, O = 16.0, Mg = 24.3, Ca = 40.1, Br = 79.9; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)



Class practice 12.3

1. Complete the following table by filling in the appropriate numbers.

	Substance	Molar mass / g mol ⁻¹	Mass / g	No. of moles / mol	No. of formula units
(a)	Sodium hydroxide			0.250	
(b)	Helium		0.20		
(c)	Sulphur dioxide				3.01×10^{24}
(d)	Compound X		23.0		3.01×10^{23}

(Relative atomic masses: H = 1.0, He = 4.0, O = 16.0, Na = 23.0, S = 32.1; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

2. Which of the following gases contains the greatest number of molecules?

(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0, F = 19.0, Cl = 35.5, Ar = 40.0)

- A. 50.0 g of argon
B. 50.0 g of fluorine
C. 50.0 g of hydrogen chloride
D. 50.0 g of nitrogen monoxide



3. Calculate the mass of chlorine which contains the same number of molecules as there are in 1.20 mol of water.

12.3 Calculating percentage by mass of an element in a compound

From the formula of a compound, we can work out the **percentage by mass** of each element in the compound.



Key point

Percentage by mass of element A in a compound

$$= \frac{\text{relative atomic mass of } A \times \text{number of atoms of } A \text{ in the formula}}{\text{formula mass of the compound}} \times 100\%$$



Example 12.4

Calculating the percentage by mass of an element in a compound

Bauxite is an ore of aluminium. It contains mainly aluminium oxide (Al_2O_3). Calculate the percentage by mass of aluminium in aluminium oxide.

(Relative atomic masses: $\text{O} = 16.0$, $\text{Al} = 27.0$)

Solution

Formula mass of Al_2O_3

$$\begin{aligned} & \text{Al}_2 \quad \text{O}_3 \\ &= 27.0 \times 2 + 16.0 \times 3 \\ &= 102.0 \end{aligned}$$

Percentage by mass of Al in Al_2O_3

$$\begin{aligned} &= \frac{\text{relative atomic mass of Al} \times \text{number of atoms of Al in the formula}}{\text{formula mass of Al}_2\text{O}_3} \times 100\% \\ &= \frac{27.0 \times 2}{102.0} \times 100\% \\ &= 52.9\% \end{aligned}$$

Self-test 12.4

Sodium hydroxide is an active ingredient of drain cleaners. Calculate the percentage by mass of sodium in sodium hydroxide.

(Relative atomic masses: $\text{H} = 1.0$, $\text{O} = 16.0$, $\text{Na} = 23.0$)



Class practice 12.4

1. Calculate the percentage by mass of sulphur in the compound Na_2S .
(Relative atomic masses: Na = 23.0, S = 32.1)
 2. Calculate the percentage by mass of nitrogen in the compound $(\text{NH}_4)_2\text{HPO}_4$.
(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0, P = 31.0)
 3. Calculate the percentage by mass of magnesium in magnesium chloride-6-water $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$.
(Relative atomic masses: H = 1.0, O = 16.0, Mg = 24.3, Cl = 35.5)
 4. Given that an oxide of iron has 69.9% of iron by mass and the formula mass of the oxide is 159.6, how many iron atoms are there in a formula unit of the oxide?
(Relative atomic mass: Fe = 55.8)
- A. 1
B. 2
C. 3
D. 4



12.4 Calculating the mass of an element in a compound

The mass of an element in a compound can be calculated from the mass of the compound and the percentage by mass of the element in that compound.



Example 12.5

Calculating the mass of an element in a compound

How many grams of aluminium can be extracted from 2000 g of aluminium oxide?

Solution

From Example 12.4, we know that the percentage by mass of Al in Al_2O_3 is 52.9%. That means for every gram of Al_2O_3 , there is 0.529 g of Al in it.

$\therefore$ mass of Al in 2000 g of Al_2O_3

$$= 2000 \text{ g} \times 52.9\%$$

$$= 1058 \text{ g}$$





Example 12.6

Calculating the mass of copper in hydrated copper(II) sulphate

Calculate the mass of copper in 15.0 g of copper(II) sulphate-5-water ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$).

(Relative atomic masses: H = 1.0, O = 16.0, S = 32.1, Cu = 63.5)

Solution

Formula mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

$$= 63.5 + 32.1 + 16.0 \times 4 + 5 \times (1.0 \times 2 + 16.0)$$

$$= 249.6$$

Percentage by mass of Cu in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

$$= \frac{\text{relative atomic mass of Cu} \times \text{number of atoms of Cu in the formula}}{\text{formula mass of } \text{CuSO}_4 \cdot 5\text{H}_2\text{O}} \times 100\%$$

$$= \frac{63.5}{249.6} \times 100\%$$

$$= 25.4\%$$

That means for every gram of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, there is 0.254 g of Cu in it.

$\therefore$ mass of Cu in 15.0 g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

$$= 15.0 \text{ g} \times 25.4\%$$

$$= 3.81 \text{ g}$$



Learning tip

Crystals of some salts contain water molecules as part of their structure. These water molecules are called water of crystallization e.g. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.
water of crystallization

Self-test 12.6

Calculate the mass of potassium in 7.91 g of potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$).

(Relative atomic masses: O = 16.0, K = 39.1, Cr = 52.0)



Class practice 12.5

1. Calculate the mass of calcium in 206 g of calcium oxide.
(Relative atomic masses: O = 16.0, Ca = 40.1)
2. Calculate the mass of oxygen in 40.0 g of carbon dioxide.
(Relative atomic masses: C = 12.0, O = 16.0)
3. A sample of sodium nitrate contains 100 g of sodium. Calculate the mass of nitrogen present in this sample.
(Relative atomic masses: N = 14.0, O = 16.0, Na = 23.0)

12.5 Calculating the relative atomic mass of an element

The relative atomic mass of an element may be calculated from the formula of the compound and percentage by mass of that element in the compound.



Example 12.7

Calculating the relative atomic mass of an element

The chloride of metal M has the formula of MCl_3 and contains 34.4% by mass of M . Calculate the relative atomic mass of M .

(Relative atomic mass: Cl = 35.5)

Solution

Let the relative atomic mass of M be a .

Percentage by mass of M in MCl_3

$$= \frac{\text{relative atomic mass of } M \times \text{number of atoms of } M \text{ in the formula}}{\text{formula mass of } MCl_3} \times 100\%$$

$$34.4\% = \frac{a}{a + 35.5 \times 3} \times 100\%$$

$$a = 55.8$$

$\therefore$ the relative atomic mass of M is 55.8.



Math tip

To solve a ,

$$0.344 = \frac{a}{a + 35.5 \times 3} \times 1$$

$$0.344a + 36.6 = a$$

$$0.656a = 36.6$$

$$a = 55.8$$

Self-test 12.7

The bromide of metal X has the formula of XBr_2 and contains 25.6% by mass of X . Calculate the relative atomic mass of X .

(Relative atomic mass: Br = 79.9)



Class practice 12.6

1. A metal oxide MO contains 79.87% by mass of M . Calculate the relative atomic mass of M .
(Relative atomic mass: O = 16.0)
2. 26.88 g of a metal chloride MCl contains 5.68 g of chlorine. Calculate the relative atomic mass of M .
(Relative atomic mass: Cl = 35.5)

12.6 Determining empirical formulae from experimental data

What is empirical formula?

In Chapter 8, we have seen the molecular formulae and structural formulae of some simple molecular substances. These types of chemical formulae are used to represent elements and covalent compounds consisting of molecules. In this section, we are going to learn another type of chemical formula, which is **empirical formula**.

The empirical formula of a compound is defined as the formula that shows the **simplest whole number ratio** of the atoms or ions present in the compound.

- For ionic compounds, the empirical formulae are the same as the chemical formulae. For example, the empirical formulae for sodium chloride and potassium carbonate are NaCl and K_2CO_3 respectively.
- For molecular substances, the empirical formulae may or may not be the same as the chemical formulae. See Section 12.7 for more details.



Key point

The **empirical formula** of a compound is the formula which shows the **simplest whole number ratio** of the atoms or ions present.

The empirical formula of a compound can be determined from its **composition by mass** i.e. the mass of each element present in a known mass of the compound. The composition of a compound has to be determined by experiments.

Experiment 12.1



Determining the empirical formula of an oxide of copper



Experiment video

Determining the empirical formula of an oxide of copper

Determining the empirical formula of an oxide of copper

In Chapter 7, we learnt that copper forms two oxides with oxygen, namely, copper(I) oxide (Cu_2O) and copper(II) oxide (CuO). To determine the empirical formula of an oxide of copper, we have to find the ratio by mass of copper and oxygen in the compound in an experiment.

A set-up for conducting the experiment is shown in Figure 12.3.

SBA note

- At the beginning of the experiment, town gas is passed into the combustion tube to expel the air inside the tube.
- The hot copper formed may react with oxygen in the air again. Therefore, it is necessary to pass town gas through the combustion tube, even after heating has stopped.

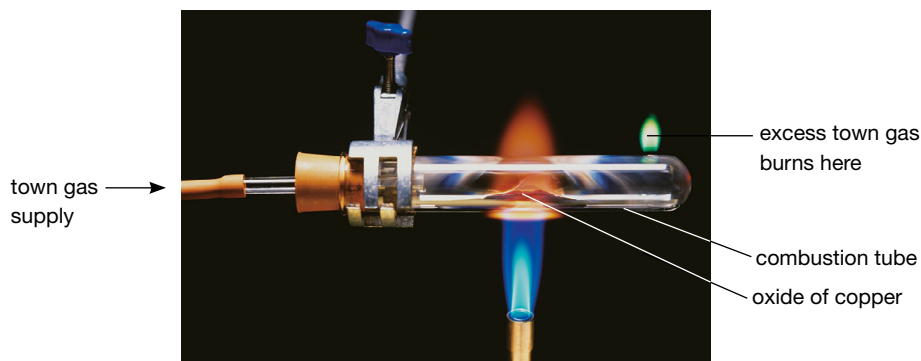
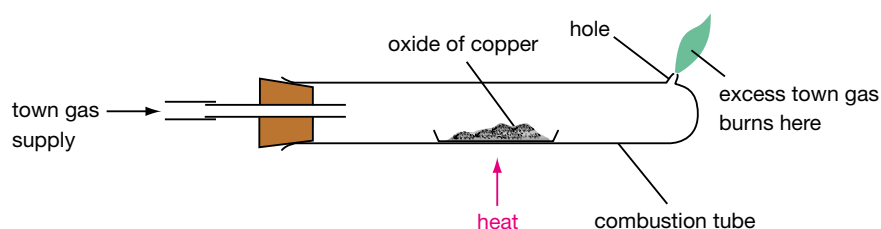


Figure 12.3 Set-up for determining the empirical formula of an oxide of copper

Learning tip

Hydrogen and carbon monoxide reduce the oxide by removing the oxygen from it.

Put a **known mass** of the oxide of copper (black) inside a combustion tube. Pass **town gas** into the tube while heating the oxide strongly (Figure 12.3). **Hydrogen** and **carbon monoxide** in the town gas **reduce** the oxide of copper to copper (reddish brown). Keep heating the solid in the combustion tube until there is no further colour change in the solid. Then find the mass of the solid (copper) remained.

The specimen results of the experiment are shown below:

Item	Mass / g
Mass of combustion tube	18.100
Mass of combustion tube + oxide of copper	18.701
Mass of combustion tube + copper	18.579
Mass of copper in the oxide	$18.579 - 18.100 = 0.479$
Mass of oxygen in the oxide	$18.701 - 18.579 = 0.122$

From the experimental results, we can work out the empirical formula of the oxide of copper as shown in 'Problem-solving strategy 12.1'.



Problem-solving strategy 12.1

Determining the empirical formula of a compound

The steps showing how to determine the empirical formula of the above oxide of copper are as follows:

		Cu	O
1 Write the mass of each element	Mass / g	0.479	0.122
2 Write the relative atomic mass of each element (After some practice, you will be able to skip this step.)	Relative atomic mass	63.5	16.0
3 Calculate the number of moles of each kind of atoms	Number of moles of atoms / mol (= $\frac{\text{mass (g)}}{\text{molar mass (g mol}^{-1}\text{)}}$)	$\frac{0.479}{63.5} = 0.00754$	$\frac{0.122}{16.0} = 0.00763$
4 Calculate the simplest whole number mole ratio of atoms	Simplest whole number mole ratio of atoms (divided by the smallest number of moles) (multiplied by the smallest possible whole number to turn all the values into whole number if necessary)	$\frac{0.00754}{0.00754} = 1$	$\frac{0.00763}{0.00754} = 1.01 \approx 1$

$\therefore$ the empirical formula of the oxide of copper is CuO.

Note:

Due to experimental error, the number(s) obtained for the simplest mole ratio when determining the empirical formulae may have a **small difference** from a whole number. It is acceptable to '**round off**' the value(s) to the nearest whole number(s). However, we must be very cautious when doing so. For example, 1.01 can be rounded off to 1, but 1.2 is usually **not** rounded off to 1.





Example 12.8

Determining the empirical formula of a compound from the percentage by mass

An oxide of lead contains 92.7% of lead by mass. Determine the empirical formula of this oxide.

(Relative atomic masses: O = 16.0, Pb = 207.2)

Solution

	Pb	O
Mass / g	92.7	7.3
Relative atomic mass	207.2	16.0
Number of moles of atoms / mol	$\frac{92.7}{207.2} = 0.447$	$\frac{7.3}{16.0} = 0.456$
Simplest whole number mole ratio of atoms	$\frac{0.447}{0.447} = 1$	$\frac{0.456}{0.447} = 1.02 \approx 1$

$\therefore$ the empirical formula of the oxide is PbO.



Learning tip

Assume that there is 100 g of X. Then there are 92.7 g of Pb and 7.3 g of O.



Flipped classroom

Determining the empirical formula of a compound from the percentage by mass

Self-test 12.8

Compound Y contains 26.95% sulphur, 13.44% oxygen and 59.61% chlorine by mass. Determine the empirical formula of Y.

(Relative atomic masses: O = 16.0, S = 32.1, Cl = 35.5)



Class practice 12.7

1. A compound has the empirical formula C_xH_y . On analysis, 1.000 g of this compound was found to contain 0.857 g of carbon. Calculate the values of x and y.
(Relative atomic masses: H = 1.0, C = 12.0)
2. A 16.28 g sample of a metal oxide, MO, upon reduction, gives 13.08 g of the metal M. Which of the following is the relative atomic mass of M?
(Relative atomic mass: O = 16.0)
A. 24.3
B. 40.1
C. 63.5
D. 65.4
3. A compound contains iron and oxygen only. In an experiment for determining the empirical formula of this compound, 2.31 g of the compound was heated with carbon monoxide. Upon complete reaction, carbon dioxide and 1.67 g of iron were formed. Calculate the empirical formula of this compound.

(HKDSE 2015 Paper 1B Q3b(ii))

12.7 Determining molecular formulae from experimental data

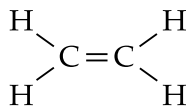


Figure 12.4 Structural formula of ethene

What is molecular formula?

Ethene is a covalent compound. It has two carbon and four hydrogen atoms in each molecule (Figure 12.4). Therefore, the empirical formula of ethene is CH_2 .

The empirical formula cannot show the **actual numbers** of carbon and hydrogen atoms in an ethene molecule. To represent their actual numbers, we have to use the **molecular formula**.



Key point

The **molecular formula** of a substance shows the **actual number** of each kind of atoms in **one** molecule of the substance.

The concept of empirical formula applies to **both ionic and covalent compounds**. The concept of molecular formula **only applies to molecular substances**. Table 12.2 shows the empirical formulae and molecular formulae of some compounds.

Compound	Empirical formula	Molecular formula
Carbon dioxide	CO_2	CO_2
Ethene	CH_2	C_2H_4
Propene	CH_2	C_3H_6
Ethanol	$\text{C}_2\text{H}_6\text{O}$	$\text{C}_2\text{H}_6\text{O}$
Glucose	CH_2O	$\text{C}_6\text{H}_{12}\text{O}_6$
Magnesium carbonate	MgCO_3	/
Silicon dioxide	SiO_2	/

Table 12.2 Empirical formulae and molecular formulae of some compounds

We should note that the empirical and molecular formulae of a compound may be the same (e.g. carbon dioxide) or different (e.g. ethene). The molecular formula is the empirical formula **multiplied by an integer** (1, 2, 3, etc.).

Determining the molecular formulae of covalent compounds

Once the empirical formula and the relative molecular mass of a compound are known, we can determine the molecular formula of the compound. See Examples 12.9–12.11.



Example 12.9

Determining molecular formula from empirical formula and relative molecular mass

A covalent compound has an empirical formula of CH_2 and a relative molecular mass of 42.0. Determine its molecular formula.

(Relative atomic masses: $\text{H} = 1.0$, $\text{C} = 12.0$)

Solution

Let the molecular formula of the compound be $(\text{CH}_2)_n$, where n is an integer.

$$n \times (12.0 + 1.0 \times 2) = 42.0$$

$$n = 3$$

$\therefore$ the molecular formula of the compound is $(\text{CH}_2)_3$, i.e. C_3H_6 .



Example 12.10

Determining empirical formula and molecular formula from percentage by mass

Compound X was found to contain 80% carbon and 20% hydrogen by mass. Given that X has a relative molecular mass smaller than 35.0, what are the empirical formula and molecular formula of X ?

(Relative atomic masses: $\text{H} = 1.0$, $\text{C} = 12.0$)

Solution

	C	H
Mass / g	80	20
Relative atomic mass	12.0	1.0
Number of moles of atoms / mol	$\frac{80}{12.0} = 6.67$	$\frac{20}{1.0} = 20.0$
Simplest whole number mole ratio of atoms	$\frac{6.67}{6.67} = 1$	$\frac{20.0}{6.67} = 3$

$\therefore$ the empirical formula of X is CH_3 .

Let the molecular formula of X be $(\text{CH}_3)_n$, where n is an integer.

$$n \times (12.0 + 1.0 \times 3) < 35.0$$

$$n < 2.33 \text{ (i.e. } n = 2\text{)}$$

$\therefore$ the molecular formula of X is C_2H_6 .



Learning tip

Assume that there is 100 g of X . Then there are 80 g of C and 20 g of H.



Math tip

The symbol ' $<$ ' means 'smaller than'.



Example 12.11

Determining empirical formula and molecular formula from masses of combustion products

Compound Z contains only carbon, hydrogen and oxygen. 2.43 g of Z burns completely to give 3.96 g of carbon dioxide and 1.35 g of water. Determine the empirical formula of Z . If the relative molecular mass of Z is 162.0, determine its molecular formula.

(Relative atomic masses: $H = 1.0$, $C = 12.0$, $O = 16.0$)

Solution

Since all the C in CO_2 and H in H_2O come from Z ,

mass of C in Z

$$= 3.96 \text{ g} \times \frac{12.0}{12.0 + 16.0 \times 2}$$

$$= 1.08 \text{ g}$$

mass of H in Z

$$= 1.35 \text{ g} \times \frac{1.0 \times 2}{1.0 \times 2 + 16.0}$$

$$= 0.15 \text{ g}$$

The rest of the mass of Z must come from oxygen.

$\therefore$ mass of O in Z

$$= (2.43 - 1.08 - 0.15) \text{ g}$$

$$= 1.20 \text{ g}$$

Now go on to find the empirical formula of Z as follows:

	C	H	O
Mass / g	1.08	0.15	1.20
Relative atomic mass	12.0	1.0	16.0
Number of moles of atoms / mol	$\frac{1.08}{12.0} = 0.090$	$\frac{0.15}{1.0} = 0.15$	$\frac{1.20}{16.0} = 0.075$
Simplest whole number mole ratio of atoms	$\frac{0.090}{0.075} = 1.2$	$\frac{0.15}{0.075} = 2$	$\frac{0.075}{0.075} = 1$
	$1.2 \times 5 = 6$	$2 \times 5 = 10$	$1 \times 5 = 5$

$\therefore$ the empirical formula of Z is $C_6H_{10}O_5$.

Let the molecular formula of Z be $(C_6H_{10}O_5)_n$, where n is an integer.

$$n \times (12.0 \times 6 + 1.0 \times 10 + 16.0 \times 5) = 162.0$$

$$n = 1$$

$\therefore$ the molecular formula of Z is $C_6H_{10}O_5$.

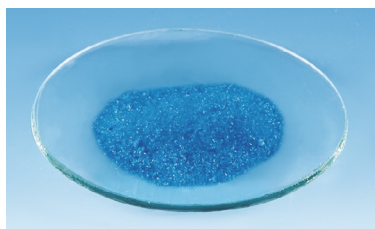




Example 12.12

Determining the number of water of crystallization in a hydrated salt

5.60 g of a hydrated copper(II) sulphate $\text{CuSO}_4 \cdot n\text{H}_2\text{O}$ is heated in a crucible to remove the water of crystallization. The white residue is anhydrous copper(II) sulphate, which is found to have a mass of 3.59 g.



Hydrated copper(II) sulphate



Anhydrous copper(II) sulphate

Experiment 12.2



Determining the formula of hydrated copper(II) sulphate



Experiment video

Determining the formula of hydrated copper(II) sulphate

- Deduce, from the experimental results, a reasonable value for n .
 - Suggest a reason why the value of n deduced in (a) differs slightly from the value calculated.
- (Relative atomic masses: H = 1.0, O = 16.0, S = 32.1, Cu = 63.5)

Solution

- Mass of water of crystallization = $(5.60 - 3.59) \text{ g} = 2.01 \text{ g}$

	CuSO_4	H_2O
Mass / g	3.59	2.01
Formula mass	159.6	18.0
Number of moles of formula units / mol	$\frac{3.59}{159.6} = 0.0225$	$\frac{2.01}{18.0} = 0.112$
Simplest whole number mole ratio of formula units	$\frac{0.0225}{0.0225} = 1$	$\frac{0.112}{0.0225} = 4.98$

Since n should be a whole number, a reasonable value of n would be 5.

- The experimental value of n (4.98) is smaller than 5. This might be because not all water of crystallization has been removed from the hydrated salt during the heating process.



Class practice 12.8

- A compound contains only carbon, hydrogen and oxygen. 0.81 g of the compound gives 1.32 g of carbon dioxide and 0.45 g of water upon complete combustion. Given that the relative molecular mass of the compound is 320.0, determine its
 - empirical formula.
 - molecular formula.
 (Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

Cont'd

2. In an experiment to determine the number of water of crystallization in a hydrated salt of magnesium sulphate, 2.50 g sample of the salt was heated gently in a crucible until there was no further change. The results are tabulated below:

Item	Mass / g
Mass of crucible	24.30
Mass of crucible and $\text{MgSO}_4 \cdot n\text{H}_2\text{O}$	26.80
Mass of crucible and salt after heating	25.52

- (a) Calculate the mass of water evaporated from the sample.
 (b) Calculate the mass of MgSO_4 .
 (c) Calculate the mole ratio of MgSO_4 to H_2O .
 (d) State the empirical formula of the hydrated MgSO_4 .
 (Relative atomic masses: H = 1.0, O = 16.0, Mg = 24.3, S = 32.1)

12.8 Reacting masses from chemical equations



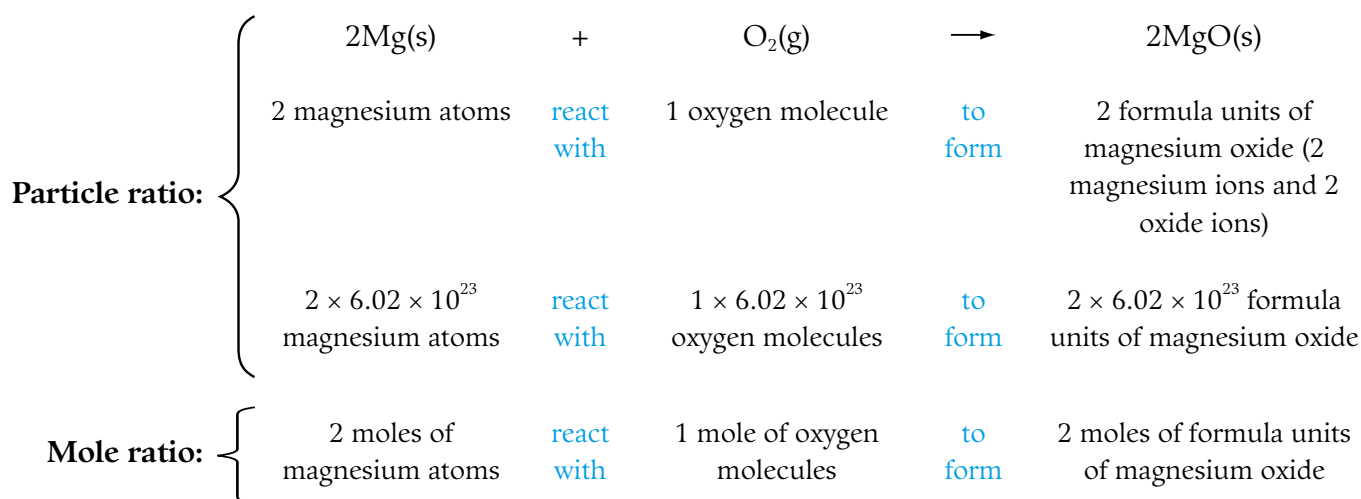
Learning tip

The quantitative study of reactants and products in a reaction is called **stoichiometry**.

Relationship between mole ratio and stoichiometric coefficients a chemical equation

In Chapter 11, we have learnt that a balanced chemical equation provides us information about the **relative numbers of particles of reactant(s) and the product(s)** taking part in a reaction.

Consider the reaction between magnesium and oxygen:



The **stoichiometric coefficients** in the equation indicate the **mole ratio** of reactants and products in the reaction. In the above reaction, the mole ratio is 2 : 1 : 2 for Mg to O₂ to MgO.

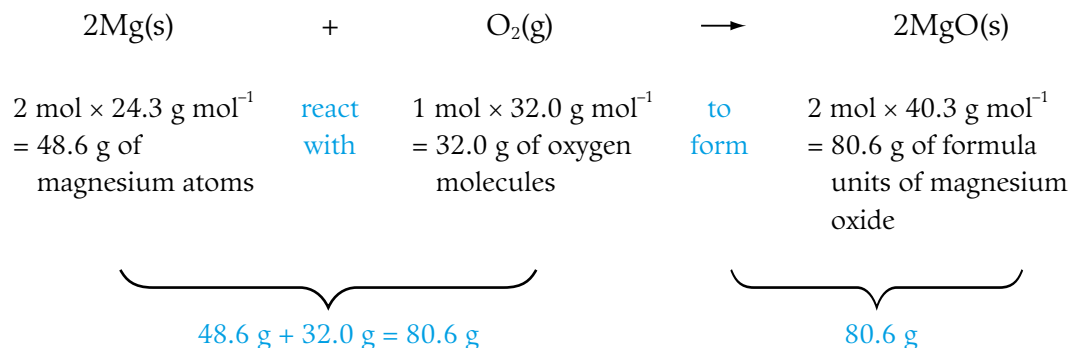
Conservation of mass

Besides the mole ratio, the chemical equation also tells us information about the reacting masses:



Think about it

How do we calculate the molar masses of Mg, O₂ and MgO respectively?



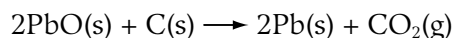
From the above calculation, the total mass of the products is equal to the total mass of the reactants. This is called **conservation of mass**. Mass is conserved because the numbers of the different kinds of atoms remain unchanged before and after a reaction.



Example 12.13

Understanding conservation of mass

Lead(II) oxide reacts with carbon powder to give lead and carbon dioxide.



w grams of PbO(s) react with x grams of C(s) to give y grams of Pb(s) and z grams of CO₂(g). Express y in terms of w , x and z .

Solution

Total mass of the reactants at the start = $(w + x)$ grams

Total mass of the products at the end = $(y + z)$ grams

According to the conservation of mass, $w + x = y + z$

$$\therefore y = w + x - z$$



Learning tip

The stoichiometric coefficients in the equation shows the quantitative relationship (i.e. mole ratio), but not the mass relationship among the species in a reaction.

Calculating reacting masses from chemical equations

Chemists often want to know the amount of product(s) that can be produced from a given amount of reactant(s). See Problem-solving strategy 12.2.



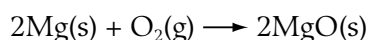
Problem-solving strategy 12.2

Calculating reacting masses from chemical equations

Calculate the mass of magnesium oxide formed when 2.43 g of magnesium burns completely in air.

(Relative atomic masses: O = 16.0, Mg = 24.3)

- 1 Write the chemical equation for the reaction.



- 2 Convert the mass(es) of the given substance(s) into the number of moles.

Number of moles of Mg (mol)

$$= \frac{\text{mass of Mg (g)}}{\text{molar mass of Mg (g mol}^{-1}\text{)}}$$

$$= \frac{2.43 \text{ g}}{24.3 \text{ g mol}^{-1}}$$

$$= 0.100 \text{ mol}$$

- 3 Calculate the number of moles of the substance asked in the question.

From the equation, mole ratio of Mg to MgO = 2 : 2 (or simply 1 : 1)

$\therefore$ number of moles of MgO formed = 0.100 mol

- 4 Convert the number of moles of that substance into mass.

Mass of MgO (g)

$$= \text{number of moles of MgO (mol)} \times \text{molar mass of MgO (g mol}^{-1}\text{)}$$

$$= 0.100 \text{ mol} \times (24.3 + 16.0) \text{ g mol}^{-1}$$

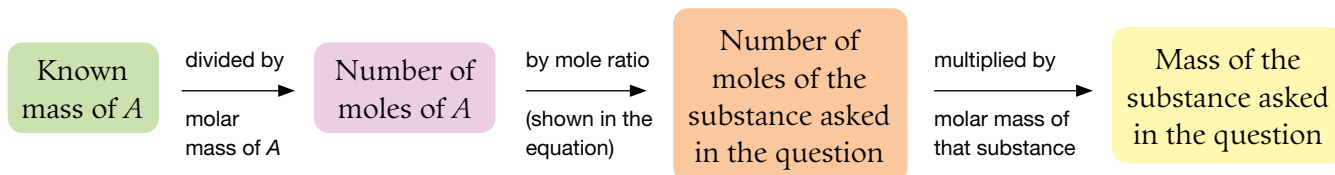
$$= 4.03 \text{ g}$$

Try it now

Calculate the mass of copper formed when 15.9 g of copper(II) oxide reacts completely with hydrogen.

(Relative atomic masses: O = 16.0, Cu = 63.5)

The flow chart below illustrates the steps for calculating the reacting masses from a chemical equation.



(Note: *A* represents the chemical formula of a given substance.)



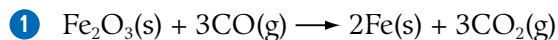
Example 12.14

Calculating the mass of a product from the mass of a reactant

Calculate the mass of iron formed when 1.60 g of iron(III) oxide reacts completely with excess carbon monoxide.

(Relative atomic masses: O = 16.0, Fe = 55.8)

Solution



② Number of moles of Fe_2O_3 used

$$= \frac{1.60 \text{ g}}{(55.8 \times 2 + 16.0 \times 3) \text{ g mol}^{-1}}$$

$$= 0.0100 \text{ mol}$$

③ From the equation, mole ratio of Fe_2O_3 to Fe = 1 : 2.

$\therefore$ number of moles of Fe

$$= 0.0100 \times 2 \text{ mol}$$

$$= 0.0200 \text{ mol}$$

④ Mass of Fe formed

$$= 0.0200 \text{ mol} \times 55.8 \text{ g mol}^{-1}$$

$$= 1.12 \text{ g}$$

Self-test 12.14

Calculate the mass of lead produced when 2.23 g of lead(II) oxide reacts completely with excess carbon.

(Relative atomic masses: O = 16.0, Pb = 207.2)

When the mass of the product of a reaction is known, we can calculate the mass of the reactant(s) based on the balanced chemical equation. See Example 12.15.



Example 12.15

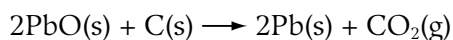
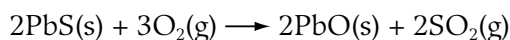
Calculating the mass of a reactant from the mass of a product

A sample of lead ore contains 75% by mass of lead(II) sulphide and 25% impurities. It is known that there are no other lead-containing compounds in the impurities. Calculate the mass of the ore required to extract 20.72 g of lead.

(Relative atomic masses: S = 32.1, Pb = 207.2)

Solution

- 1 The extraction of lead involves two steps:



- 2 Number of moles of Pb to be extracted

$$\begin{aligned} &= \frac{20.72 \text{ g}}{207.2 \text{ g mol}^{-1}} \\ &= 0.1000 \text{ mol} \end{aligned}$$

- 3 From the equation, mole ratio of Pb to PbO to PbS = 1 : 1 : 1.

$\therefore$ number of moles of PbS required = 0.1000 mol

- 4 Mass of PbS required = $0.1000 \text{ mol} \times (207.2 + 32.1) \text{ g mol}^{-1} = 23.93 \text{ g}$

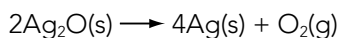
As the ore contains 75% by mass of PbS, mass of the ore required

$$\begin{aligned} &= \frac{23.93 \text{ g}}{75\%} \\ &= 31.91 \text{ g} \end{aligned}$$



Class practice 12.9

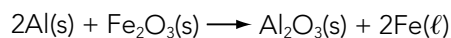
1. Upon strong heating, silver oxide decomposes to silver and oxygen:



Calculate the mass of silver obtained when 6.96 g of silver oxide is strongly heated in air.

(Relative atomic masses: O = 16.0, Ag = 107.9)

2. Aluminium powder reacts with finely divided iron(III) oxide according to the following equation:

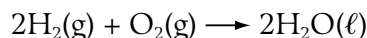


Calculate the mass of finely divided iron(III) oxide needed to produce 10.55 g of iron.

(Relative atomic masses: O = 16.0, Fe = 55.8)

Calculations involving limiting reactant

Consider the reaction between hydrogen and oxygen to form water:



The equation tells us that 1 molecule of oxygen is required to react with 2 molecules of hydrogen for complete reaction. See Figure 12.5.

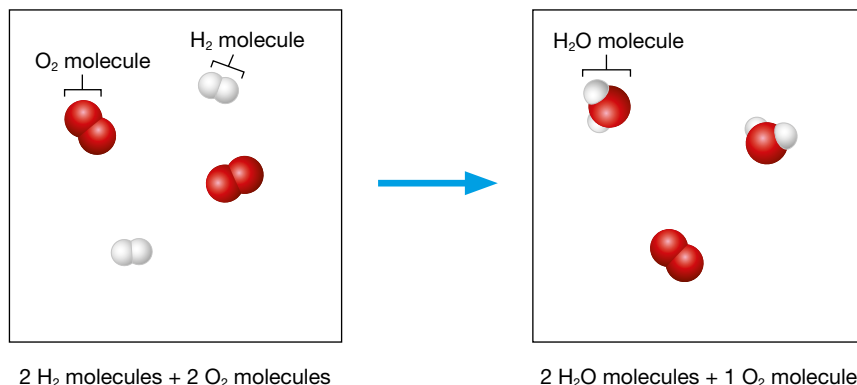


Figure 12.5 Two hydrogen molecules require one oxygen molecule for complete reaction. Therefore, oxygen is in excess.

 **Flipped classroom**
Calculations involving limiting reactant

In this reaction, all the hydrogen has reacted. Oxygen is **in excess**. The amount of water produced is **limited** by the amount of hydrogen used. Therefore, hydrogen is called the **limiting reactant** in this case. The limiting reactant **limits the amount of the product formed** in a reaction. Besides, the **reaction stops when the limiting reactant is used up**.



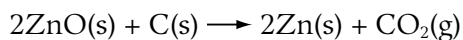
Problem-solving strategy 12.3

Determining the limiting reactant in a reaction

In the extraction of zinc from zinc oxide, 8.14 g of zinc oxide is heated with 6.50 g of carbon powder. Which reactant is the limiting reactant?

(Relative atomic masses: C = 12.0, O = 16.0, Zn = 65.4)

- 1 Write the chemical equation for the reaction.



- 2 Calculate the number of moles of each reactant.

Number of moles of ZnO

$$= \frac{8.14 \text{ g}}{(65.4 + 16.0) \text{ g mol}^{-1}} = 0.100 \text{ mol}$$

Number of moles of C

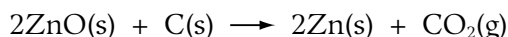
$$= \frac{6.50 \text{ g}}{12.0 \text{ g mol}^{-1}} = 0.542 \text{ mol}$$

Try it now

In an experiment, 6.54 g of zinc is heated with 3.84 g of oxygen. Which one is the limiting reactant?

Cont'd

- 3 Calculate, if all of a reactant is used up, the number of moles of the other reactant that will be needed for complete reaction.



0.100 mol 0.542 mol
(from 2) (from 2)

From the equation, mole ratio of ZnO to C is 2 : 1.

- (a) 0.100 mol of ZnO requires $\frac{0.100}{2}$ mol = 0.0500 mol of C for complete reaction
(b) 0.542 mol of C requires $0.542 \text{ mol} \times 2 = 1.084$ mol of ZnO for complete reaction

- 4 Identify the reactant that is in limited amount i.e. not in excess. This reactant will first be used up in the reaction.

In 3(a), only 0.0500 mol of C is needed to react with all of the ZnO (i.e. 0.100 mol). As there is 0.542 mol of C available, C is in excess. Thus ZnO is the limiting reactant.



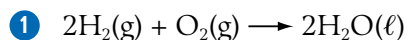
Example 12.16

Calculating reacting masses involving limiting reactant

Calculate the mass of water formed when 5.4 g of hydrogen is heated with 3.2 g of oxygen.

(Relative atomic masses: H = 1.0, O = 16.0)

Solution



2 Number of moles of $\text{H}_2 = \frac{5.4 \text{ g}}{(1.0 \times 2) \text{ g mol}^{-1}} = 2.7 \text{ mol}$

Number of moles of $\text{O}_2 = \frac{3.2 \text{ g}}{(16.0 \times 2) \text{ g mol}^{-1}} = 0.10 \text{ mol}$

- 3 From the equation, mole ratio of H_2 to $\text{O}_2 = 2 : 1$.

0.10 mol of O_2 requires only $0.10 \text{ mol} \times 2 = 0.20 \text{ mol}$ of H_2 for complete reaction.

$\therefore \text{H}_2$ is in excess while O_2 is the limiting reactant.

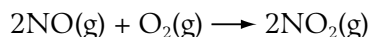
From the equation, mole ratio of O_2 to $\text{H}_2\text{O} = 1 : 2$.

Number of moles of H_2O formed = $0.10 \text{ mol} \times 2 = 0.20 \text{ mol}$

Mass of H_2O formed = $0.20 \text{ mol} \times (1.0 \times 2 + 16.0) \text{ g mol}^{-1} = 3.6 \text{ g}$

Self-test 12.16

Calculate the mass of nitrogen dioxide formed when 26.58 g of nitrogen monoxide reacts with 8.06 g of oxygen according to the following equation:



(Relative atomic masses: N = 14.0, O = 16.0)



Concept check

- ✗ The reactant which is in the least number of moles must be the limiting reactant.
- ✓ A limiting reactant is NOT necessarily the one which is in the least number of moles. It is the one that is not in excess and is first used up in a reaction.

Theoretical yield, actual yield and percentage yield

Theoretical yield is the maximum amount of product **expected** if the reaction proceeds exactly as shown in the chemical equation. **Actual yield** is the amount of product **actually** obtained at the end of the reaction. The actual yield of a reaction is often **smaller than** the theoretical yield because:

- the reaction may not complete.
- impurities are present in the reactants i.e. the actual mass of the reactant(s) used is smaller than expected.
- there are side reactions which give unwanted side products.
- some product is lost in some of the experimental processes, such as purification.

The **efficiency** of a chemical reaction can be expressed by the **percentage yield**. The higher the percentage yield, the more efficient is the chemical reaction.



Key point

$$\text{Percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$



Example 12.17

Calculating the theoretical yield and actual yield of a product

In an experiment, 31.8 g of copper(II) oxide was heated with 6.0 g of carbon powder.

- Calculate the theoretical yield of copper.
 - Given that the percentage yield of copper is 82%, calculate the actual yield of copper.
- (Relative atomic masses: C = 12.0, O = 16.0, Cu = 63.5)

Solution

- $2\text{CuO(s)} + \text{C(s)} \rightarrow 2\text{Cu(s)} + \text{CO}_2\text{(g)}$
- Number of moles of CuO = $\frac{31.8 \text{ g}}{(63.5 + 16.0) \text{ g mol}^{-1}} = 0.400 \text{ mol}$
 Number of moles of C = $\frac{6.0 \text{ g}}{12.0 \text{ g mol}^{-1}} = 0.50 \text{ mol}$

Cont'd

- 3 From the equation, mole ratio of CuO to C = 2 : 1.

0.400 mol of CuO only requires $\frac{0.400}{2}$ mol = 0.200 mol of C for complete reaction.

$\therefore$ C is in excess while CuO is the limiting reactant.

From the equation, mole ratio of CuO to Cu = 1 : 1.

$\therefore$ number of moles of Cu = 0.400 mol

Theoretical yield of Cu = 0.400 mol $\times$ 63.5 g mol⁻¹ = 25.4 g

- (b) Actual yield of Cu (g) = theoretical yield of Cu (g) $\times$ percentage yield of Cu (%)
 = 25.4 g $\times$ 82% = 20.8 g



Example 12.18

Integrated

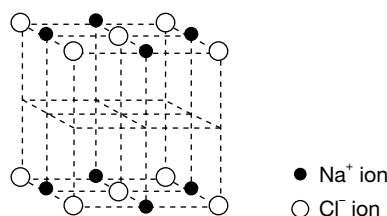
Calculating the total mass of ions in a cube of sodium chloride crystal

Sodium chloride is one of the most abundant minerals on the Earth. It can be extracted from sea water or some ores in the Earth's crust. Crystals of sodium chloride have a giant ionic structure.



Sodium chloride crystals

- (a) Outline how to obtain crystals of sodium chloride from a muddy sea water sample in the school laboratory.
- (b) Suggest a method for extracting sodium from sodium chloride.
- (c) The diagram below shows a part of the structure of sodium chloride crystal with some ions missing.



Complete the diagram by filling in the missing ions.

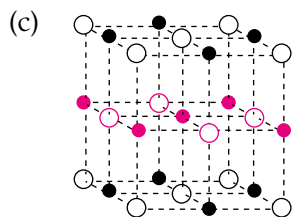
- (d) From an experiment, it is found that there are 4 Na⁺ ions and 4 Cl⁻ ions in a cube of sodium chloride crystal, and the length of each side of the cube is 5.65×10^{-8} cm.
- (i) Express the total mass of 4 Na⁺ ions and 4 Cl⁻ ions in terms of the Avogadro constant, L.
 (Relative atomic masses: Na = 23.0, Cl = 35.5)
- (ii) Calculate the volume of the cube of sodium chloride crystal.
- (iii) Hence, determine the value of the Avogadro constant, L, given that the density of sodium chloride is 2.17 g cm⁻³.
- (e) Refer to the following chemical reaction:
- $$2\text{Na(s)} + \text{Cl}_2\text{(g)} \longrightarrow 2\text{NaCl(s)}$$
- n moles of Na(s) are allowed to react with $2n$ moles of Cl₂(g) under suitable conditions until the reaction stops. Calculate the number of moles of NaCl(s) formed in terms of n .

Cont'd

Solution

(a) Filter the sea water sample to get rid of the mud. Then heat the filtered sea water until it becomes saturated. After that, allow the hot saturated sea water to cool slowly to room temperature. Crystals of sodium chloride will form gradually. Finally, separate the crystals of sodium chloride from the remaining sea water by filtration.

(b) Electrolysis of the molten sodium chloride



Learning tip
In part (a), the crystals obtained are **not** completely pure. They also contain other salts e.g. CaCl_2 , MgSO_4 , etc.

(d) (i) Mass of 1 Na^+ ion = $\frac{23.0 \text{ g mol}^{-1}}{L \text{ mol}^{-1}}$; mass of 1 Cl^- ion = $\frac{35.5 \text{ g mol}^{-1}}{L \text{ mol}^{-1}}$

Total mass of 4 Na^+ ions and 4 Cl^- ions

$$= 4 \times \frac{23.0 \text{ g mol}^{-1}}{L \text{ mol}^{-1}} + 4 \times \frac{35.5 \text{ g mol}^{-1}}{L \text{ mol}^{-1}} = \frac{234}{L} \text{ g}$$

(ii) Volume of the cube of sodium chloride crystal

$$= (5.65 \times 10^{-8} \text{ cm})^3 = 1.80 \times 10^{-22} \text{ cm}^3$$

(iii) Mass of the cube of NaCl crystal

$$= 2.17 \text{ g cm}^{-3} \times 1.80 \times 10^{-22} \text{ cm}^3 = 3.906 \times 10^{-22} \text{ g}$$

Mass of the cube of NaCl crystal

= total mass of 4 Na^+ ions and 4 Cl^- ions in the cube of crystal

$$\therefore \frac{234}{L} \text{ g} = 3.906 \times 10^{-22} \text{ g}$$

$$L = 5.99 \times 10^{23}$$

(e) From the given equation, mole ratio of Na to Cl_2 to NaCl = 2 : 1 : 2.

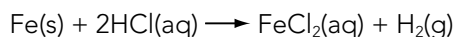
$\therefore n$ moles of Na requires only $\frac{n}{2}$ moles of Cl_2 for complete reaction. Hence, Na(s) is the limiting reactant.

$\therefore$ number of moles of NaCl formed = n moles



Class practice 12.10

1. Iron reacts with dilute hydrochloric acid according to the following equation:



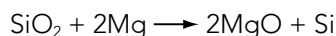
0.5 mol of iron is added to an aqueous solution containing 0.8 mol of hydrochloric acid. Which of the following statements is correct?

- A. 0.5 mol of hydrogen forms.
- B. 0.8 mol of iron(II) chloride forms.
- C. Iron is in excess by 0.1 mol.
- D. Hydrochloric acid is in excess by 0.3 mol.



Cont'd

2. Under certain conditions, 1.0 g of SiO_2 is allowed to react with 1.0 g of Mg. The equation for the reaction is shown below:



Calculate the theoretical mass of Si that can be formed.

(Relative atomic masses: O = 16.0, Mg = 24.3, Si = 28.1)

(HKDSE 2021 Paper 1B Q3c(iii))

3. A student performed an experiment to prepare calcium hydroxide. 1.50 g of calcium granules was added to a sufficient amount of water. The calcium hydroxide precipitate formed was then filtered off, washed and dried.
- Write the chemical equation for the reaction of calcium with water.
 - Calculate the theoretical yield of calcium hydroxide.
 - The actual yield of calcium hydroxide was much smaller than the theoretical yield. Suggest a reason why there was such a difference.
- (Relative atomic masses: H = 1.0, O = 16.0, Ca = 40.1)

Activity 12.1



Determining the chemical reaction occurring in the thermal decomposition of baking soda

Sodium hydrogencarbonate (commonly called baking soda) is a main ingredient of baking powder. Baking powder is used in bread-making to ensure that the bread can 'rise' during baking. When the temperature of the bread mixture reaches about 50°C , the sodium hydrogencarbonate (NaHCO_3) in the powder decomposes and carbon dioxide evolves.

Sodium hydrogencarbonate is proposed to have undergone the following three decomposition reactions during the baking process. In fact, only one of these reactions actually occurs.

- sodium hydrogencarbonate $\rightarrow$ sodium hydroxide + carbon dioxide
- sodium hydrogencarbonate $\rightarrow$ sodium oxide + carbon dioxide + water
- sodium hydrogencarbonate $\rightarrow$ sodium carbonate + carbon dioxide + water



 Baking powder

Experiment 12.3



Studying the thermal decomposition of sodium hydrogencarbonate



Experiment video

Studying the thermal decomposition of sodium hydrogencarbonate

Tasks

Form groups of two to answer the following questions.

- Transcribe the above word equations into chemical equations. (Hint: except carbon dioxide and water, all substances are solids in these reactions.)
- To identify the chemical reaction that has taken place during the baking process, 2.53 g of sodium hydrogencarbonate was heated until there was no further changes. It was found that 1.59 g of a solid remained at the end of the experiment. Which of the above reactions matches with the result?
(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0, Na = 23.0)

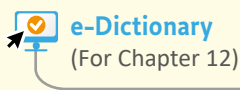
Chapter review

Progress check

After studying this chapter, you should be able to:

- ☒ perform calculations related to moles, Avogadro constant and molar masses
- ☒ calculate the percentage by mass of an element or the mass of an element in a compound using appropriate information
- ☒ calculate the relative atomic mass of an element using appropriate information
- ☒ determine empirical formulae and molecular formulae from composition by mass and molar masses
- ☒ understand the quantitative relationship of the reactants and products in a reaction as revealed by a chemical equation
- ☒ calculate masses of reactants and products in a reaction from the relevant equation and state the interrelationship between them
- ☒ solve problems involving limiting reagents

Key terms



	Page		Page
1. actual yield 實際產量	92	7. mole 摩爾	67
2. Avogadro constant 亞佛加德羅常數	67	8. molecular formula 分子式	81
3. composition by mass 質量組成	77	9. percentage by mass 質量百分比	73
4. empirical formula 實驗式	77	10. percentage yield 百分產率	92
5. limiting reactant 限量反應物	90	11. theoretical yield 理論產量	92
6. molar mass 摩爾質量	69		

Key equations

- Number of moles (mol) = $\frac{\text{mass (g)}}{\text{molar mass (g mol}^{-1}\text{)}}$ (Section 12.2, p.71)
- Number of moles (mol) = $\frac{\text{number of formula units}}{\text{Avogadro constant (mol}^{-1}\text{)}}$ (Section 12.2, p.71)
- Percentage by mass** of element A in a compound
 = $\frac{\text{relative atomic mass of A} \times \text{number of atoms of A in the formula}}{\text{formula mass of the compound}} \times 100\%$ (Section 12.3, p.73)
- Percentage yield** is a measure of the efficiency of a chemical reaction.
 Percentage yield = $\frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$ (Section 12.8, p.92)

Key concepts

12.1 The mole and Avogadro constant

1. Chemists use **mole** (abbreviation: mol) to describe the quantity of particles in a substance.
2. The **Avogadro constant** (symbol: L) is the number of atoms in exactly 12.0 g of carbon-12, i.e. 6.02×10^{23} . Avogadro constant is equal to $6.02 \times 10^{23} \text{ mol}^{-1}$.

12.2 The mole and molar mass

3. The **molar mass** of a substance is the mass of one mole of the substance. The unit of molar mass is g mol^{-1} .

12.3 Calculating percentage by mass of an element in a compound

4. **Percentage by mass** of element A in a compound
$$= \frac{\text{relative atomic mass of } A \times \text{number of atoms of } A \text{ in the formula}}{\text{formula mass of the compound}} \times 100\%$$

12.4 Calculating the mass of an element in a compound

5. The mass of an element in a compound can be calculated from the mass of the compound and the percentage by mass of the element in that compound.

12.5 Calculating the relative atomic mass of an element

6. The relative atomic mass of an element can be calculated from the formula of the compound and percentage by mass of that element in the compound.

12.6 Determining empirical formulae from experimental data

7. The **empirical formula** of a compound is the formula which shows the simplest whole number ratio of the atoms or ions present.
8. The empirical formula of a compound can be determined from its **composition by mass**. The composition of a compound has to be determined by experiments.

12.7 Determining molecular formulae from experimental data

9. The **molecular formula** of a substance shows the actual number of each kind of atoms in one molecule of the substance.
10. The molecular formula of a compound can be determined from its empirical formula and relative molecular mass.

12.8 Reacting masses from chemical equations

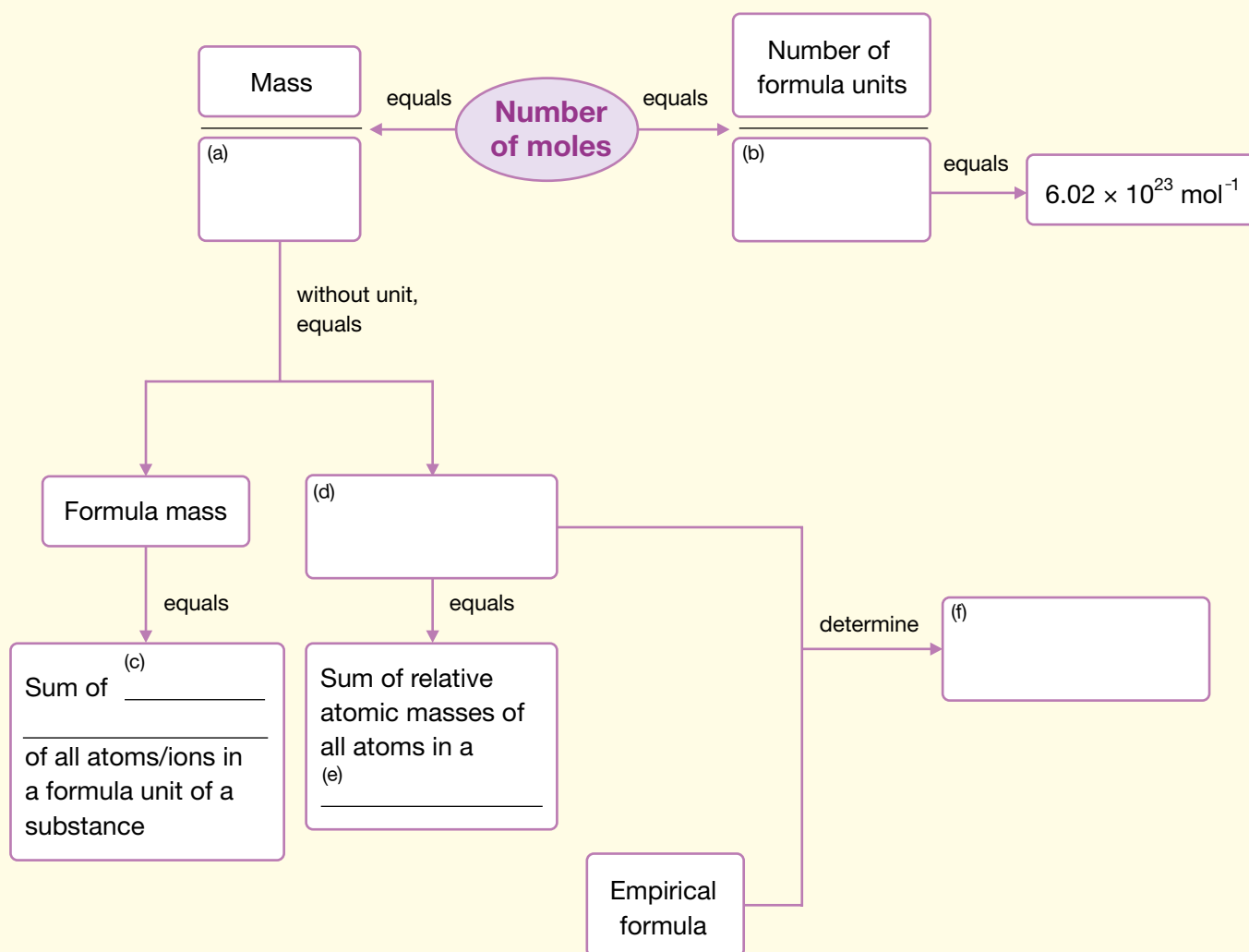
11. The total mass of the products is equal to the total mass of the reactants. This is the conservation of mass.
12. The **limiting reactant** is the reactant that is completely used up in a reaction. It limits the amount of the product(s) formed in a reaction.

13. **Theoretical yield** is the maximum amount of product expected if the reaction proceeds exactly as shown in the chemical equation.
14. **Actual yield** is the amount of product actually obtained at the end of the reaction.

► Concept map

Complete the concept map by filling in the items listed below:

Avogadro constant, molar mass, molecular formula, molecule, relative atomic masses, relative molecular mass



Chapter exercise

A. Fill in the blanks

Section 12.1

1. One mole of any substance contains _____ formula units. This number is called the _____.

Section 12.2

2. The _____ of a substance is the mass in grams of one mole of the substance.

Section 12.3

3. Percentage by mass of element *A* in a compound
 = $\frac{\text{_____ of } A \times \text{_____ of atoms of } A \text{ in the formula}}{\text{Formula mass of the compound}} \times 100\%$

Section 12.8

4. _____ is the reactant that is not in excess in a reaction. The reaction stops when it is used up.

5. Percentage yield = $\frac{\text{_____}}{\text{_____}} \times 100\%$

B. Practice questions

Section 12.2

6. Calculate the number of atoms in each of the following substances.
 (a) 2 mol of sodium
 (b) 1.5 mol of carbon monoxide
 (c) 2.0 g of oxygen
 (Relative atomic mass: O = 16.0; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)
7. Calculate the number of ions in each of the following substances.
 (a) 2 mol of ammonium chloride
 (b) 0.5 mol of sodium carbonate
 (c) 24.6 g of calcium nitrate
 (Relative atomic masses: N = 14.0, O = 16.0, Ca = 40.1; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

Section 12.3

8. Calculate the percentage by mass of
 (a) carbon in methane, CH_4 .
 (b) sulphur in sodium sulphate, Na_2SO_4 .
 (c) oxygen in iron(II) sulphate-7-water, $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$.
 (d) water in sodium carbonate-10-water, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$.
 (Relative atomic masses: H = 1.0, C = 12.0, O = 16.0, Na = 23.0, S = 32.1, Fe = 55.8)

Section 12.4

9. Calculate the mass of

- hydrogen in 10.0 g of methane, CH_4 .
- magnesium in 28.5 g of magnesium ethanoate, $(\text{CH}_3\text{COO})_2\text{Mg}$.
- chlorine in 2 mol of iron(III) chloride-6-water, $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$.
- water in 1.25 mol of calcium chloride-6-water, $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$.

(Relative atomic masses: $\text{H} = 1.0$, $\text{C} = 12.0$, $\text{O} = 16.0$, $\text{Mg} = 24.3$, $\text{Cl} = 35.5$, $\text{Ca} = 40.1$, $\text{Fe} = 55.8$)

Section 12.6

10. Determine the empirical formulae of compounds having the following composition by mass:

- 75% carbon, 25% hydrogen
- 86.6% lead, 13.4% oxygen
- 36.5% sodium, 25.4% sulphur, 38.1% oxygen
- 40.67% carbon, 23.73% nitrogen, 27.13% oxygen, 8.47% hydrogen
- A hydrated salt containing 37.11% copper, 41.68% chlorine (the rest being water of crystallization)

(Relative atomic masses: $\text{H} = 1.0$, $\text{C} = 12.0$, $\text{N} = 14.0$, $\text{O} = 16.0$, $\text{Na} = 23.0$, $\text{S} = 32.1$, $\text{Cl} = 35.5$, $\text{Cu} = 63.5$, $\text{Pb} = 207.2$)

C. Multiple-choice questions

Section 12.1

11. Which of the following substances contains the least number of ions?

- 1 mol of silver sulphide, Ag_2S
- 1 mol of copper(II) dichromate, CuCr_2O_7
- 1 mol of aluminium sulphate, $\text{Al}_2(\text{SO}_4)_3$
- 1 mol of magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$

☐

Section 12.2

12. One mole of oxygen and one mole of nitrogen have the same

- mass.
- number of atoms.
- number of molecules.

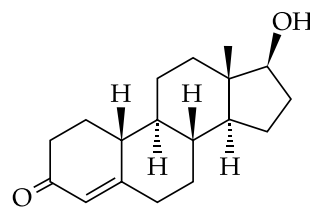
- (1) only
- (2) only
- (1) and (3) only
- (2) and (3) only

☐

Section 12.6

Direction: Questions 13 and 14 are about nandrolone.

13. Nandrolone is a drug banned by the International Olympic Committee. It can improve the performance of athletes by increasing the oxygen-carrying ability of their blood. Nandrolone has the molecular formula $\text{C}_{18}\text{H}_{26}\text{O}_2$.



Which of the following is the empirical formula of nandrolone?

- $\text{C}_9\text{H}_{13}\text{O}$
- $\text{C}_{18}\text{H}_{26}\text{O}_2$
- $\text{C}_9\text{H}_{26}\text{O}_2$
- $\text{C}_{18}\text{H}_{13}\text{O}$

☐

14. Which of the following is the molar mass of nandrolone?

(Relative atomic masses: $\text{H} = 1.0$, $\text{C} = 12.0$, $\text{O} = 16.0$)

- 137.0 g mol^{-1}
- 166.0 g mol^{-1}
- 245.0 g mol^{-1}
- 274.0 g mol^{-1}

☐

15. An oxide of metal X contains 83.0% of X . What is the formula of the oxide?

(Relative atomic masses: $O = 16.0$, $X = 39.1$)

- A. XO_2
 B. X_2O
 C. X_2O_3
 D. X_3O_2

☐

16. In an experiment, 4.40 g of a sample of $ZnSO_4 \cdot xH_2O$ was heated and 2.47 g of anhydrous $ZnSO_4$ was obtained. What is the value of x ?

(Relative atomic masses: $H = 1.0$, $O = 16.0$, $S = 32.1$, $Zn = 65.4$)

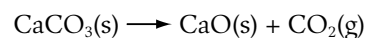
- A. 2
 B. 5
 C. 7
 D. 10

☐

Section 12.8

Direction: Questions 17 and 18 are about thermal decomposition of calcium carbonate.

17. Calcium carbonate decomposes on heating to give calcium oxide and carbon dioxide.



What would be the theoretical mass of calcium oxide produced when 10.01 g of calcium carbonate decomposes completely by heat?

(Relative atomic masses: $C = 12.0$, $O = 16.0$, $Ca = 40.1$)

- A. 4.84 g
 B. 5.61 g
 C. 7.21 g
 D. 9.62 g

☐

18. If 4.31 g of calcium oxide is finally obtained, what is the percentage yield of the reaction?

- A. 44.8%
 B. 59.8%
 C. 76.8%
 D. 89.0%

☐

D. Structured questions

Section 12.2

19. For each of the following statements, decide whether or not it is right. If it is wrong, you are required to correct it.

- (a) 32.0 g of oxygen (O_2) contains the same number of molecules as 32.0 g of ozone (O_3). (1 mark)
 (b) 1 mol of $Pb(NO_3)_2$ contains 1 mol of ions. (1 mark)

Section 12.4

20. Fluoride ions, in the form of tin(II) fluoride, are often added to toothpastes.

- (a) What is the chemical formula of tin(II) fluoride? (1 mark)
 (b) Calculate the formula mass of tin(II) fluoride. (1 mark)
 (c) Calculate the percentage by mass of fluorine in tin(II) fluoride. (1 mark)
 (d) A tube of toothpaste contains 1.50 g of tin(II) fluoride. Calculate the mass of fluorine in this tube of toothpaste. (1 mark)

(Relative atomic masses: $F = 19.0$, $Sn = 118.7$)



 Fluoride has long been used in toothpastes to prevent tooth decay.



Learn more: Building a better answer (see p.153–154)

Exam practice

A. Multiple-choice questions

Section 12.2

1. 0.25 moles of a gas has a mass of 7 g. Which of the following could be the molecular formula for the gas?
- A. C_2H_6
 B. C_2H_4
 C. C_3H_8
 D. C_3H_6

☐

(Copyright © Scottish Qualifications Authority
 National 5 2017 May Section 1 Q8)

2. Which of the following statements about 36 g of pure water is/are correct?

(Relative atomic masses: H = 1.0, O = 16.0, Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

- (1) It contains 4 mol of hydrogen atoms.
 (2) It contains $3 \times 6.02 \times 10^{23}$ of atoms.
 (3) It contains $3 \times 6.02 \times 10^{23}$ of molecules.

- A. (1) only
 B. (2) only
 C. (1) and (3) only
 D. (2) and (3) only

☐

3. If 2 g of helium contains y molecules, how many molecules are present in 38 g of fluorine?

(Relative atomic masses: He = 4.0, F = 19.0)

- A. $\frac{1}{2}y$
 B. y
 C. $2y$
 D. $4y$

☐

4. If 8.0 g of sulphur dioxide gas contains n molecules, how many molecules does 2.0 g of oxygen gas contain?

(Relative atomic masses: O = 16.0, S = 32.0)

- A. $2.0 n$
 B. $4.0 n$
 C. $0.25 n$
 D. $0.50 n$

☐

(HKDSE 2018 Paper 1A Q4)

Section 12.3

5. Which calcium compound contains the **greatest** percentage by mass of calcium?

- A. calcium carbonate
 B. calcium nitrate
 C. calcium hydroxide
 D. calcium sulfate

☐

(OCR A Level Chemistry A 2017 Jun H432/01 Q4)

Section 12.4

6. How many grams of iron can be extracted from 100 g of iron ore that contains 70% by mass of Fe_2O_3 ? (Assume that the impurities do not contain iron.)

(Relative atomic masses: O = 16.0, Fe = 55.8)

- A. 12.3 g
 B. 24.5 g
 C. 48.9 g
 D. 97.8 g

☐

7. What is the mass of oxygen in 24.0 g of $CuSO_4 \cdot 5H_2O(s)$?

(Relative atomic masses: H = 1.0, O = 16.0, S = 32.1, Cu = 63.5)

- A. 6.2 g
 B. 9.6 g
 C. 13.8 g
 D. 21.7 g

☐

(HKDSE 2020 Paper 1A Q3)

Section 12.5

8. Hydrated salt $X \cdot nH_2O$ contains 62.9% of water by mass. Given that the molar mass of X is 106.0 g mol^{-1} , what is the value of n ?

(Relative atomic masses: H = 1.0, O = 16.0)

- A. 2
 B. 5
 C. 6
 D. 10

☐

9. The composition by mass of element X in the compound K_2XO_4 is 26.8%. Which of the following statements concerning X is/are correct?

(Relative atomic masses: O = 16.0, K = 39.1)

- (1) X is a transition metal.
- (2) X is an element in Group VI of the Periodic Table.
- (3) X is an element in the fourth period of the Periodic Table.

- A. (1) only
 B. (2) only
 C. (1) and (3) only
 D. (2) and (3) only



(HKDSE 2021 Paper 1A Q19)

Section 12.6

10. The relative atomic mass of metal X is 40.1. 8.42 g of an oxide of X contains 2.40 g of oxygen. What is the mole ratio of X to oxygen in the oxide?

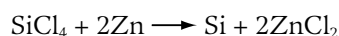
(Relative atomic mass: O = 16.0)

- A. 1 : 1
 B. 1 : 2
 C. 2 : 1
 D. 1 : 3



Section 12.8

11. Silicon can be made by heating silicon tetrachloride, $SiCl_4$, with zinc.



8.50 g of $SiCl_4$ is reacted with an excess of zinc. The percentage yield of silicon is 90%. What is the mass of silicon made?

- A. 1.26 g
 B. 1.31 g
 C. 1.40 g
 D. 1.55 g



(OCR A Level Chemistry A 2017 Jun H432/01 Paper 1 Q7)

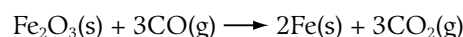
12. 0.80 g of element X is reacted with 0.40 g of O_2 to form an oxide with the formula X_2O_3 . What is the identity of element X ?

- A. Aluminium
 B. Titanium
 C. Germanium
 D. Molybdenum



(OCR A Level Chemistry A 2020 Nov H432/01 Q3)

13. Refer to the following chemical equation:



N moles of Fe_2O_3 are allowed to react with $2N$ moles of CO under suitable conditions until the reaction stops. How many moles of Fe are formed?

- A. N
 B. $2N$
 C. $\frac{2}{3}N$
 D. $\frac{4}{3}N$



(HKDSE 2014 Paper 1A Q4)

14. Zinc chloride has the formula $ZnCl_2$. Which of the following statements about zinc chloride is correct?

(Relative atomic masses: Cl = 35.5, Zn = 65.4)

- A. 1 mol of zinc atoms has been combined with 2 mol of chlorine molecules.
 B. 1 g of zinc has been combined with 2 g of chlorine.
 C. It contains a higher percentage by mass of chlorine than zinc.
 D. Its molar mass is 130.4 g mol^{-1} .



15. When 0.1 mol of an element Y is heated in air, there is an increase in mass of 2.4 g. Assume that Y reacts only with the oxygen in air and the reaction is complete. Which of the following elements might Y be?

(Relative atomic masses: C = 12.0, O = 16.0, Al = 27.0, K = 39.1, Ca = 40.1)

- A. Carbon
 B. Aluminium
 C. Potassium
 D. Calcium



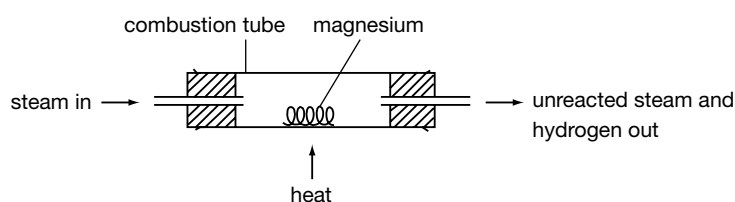
B. Structured questions

Section 12.7

16. Vanillin is a flavouring commonly used in the food industry. It has a relative molecular mass of 152.0, and contains 63.1% carbon, 5.3% hydrogen and 31.6% oxygen by mass. Deduce the molecular formula of vanillin.
(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0) (3 marks)
17. Paracetamol is the active ingredient of some commonly used painkiller. Paracetamol has a relative molecular mass of 151.0 and it has the following composition by mass: carbon 63.58%, hydrogen 5.96%, nitrogen 9.27% and oxygen 21.19%. Deduce the molecular formula of paracetamol.
(Relative atomic masses: H = 1.0, C = 12.0, N = 14.0, O = 16.0) (3 marks)

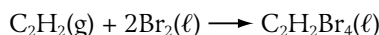
Section 12.8

18. In an experiment, steam was passed over a piece of magnesium heated in a combustion tube.



- (a) Write the chemical equation for the reaction involved. (1 mark)
- (b) Calculate the theoretical mass of magnesium oxide produced if 4.86 g of magnesium was used in the experiment. Assume that all of the magnesium had reacted. (2 marks)
- (Relative atomic masses: O = 16.0, Mg = 24.3)

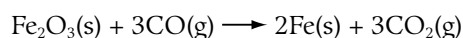
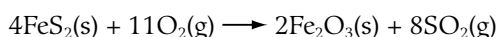
19. Ethyne (C_2H_2) reacts with bromine (Br_2) to form 1,1,2,2-tetrabromoethane ($C_2H_2Br_4$).



In an experiment, 2.00 g of ethyne reacted with 5.20 g of bromine.

- (a) Calculate the theoretical yield of $C_2H_2Br_4(l)$. (3 marks)
- (b) Given that the yield of 1,1,2,2-tetrabromoethane was 90.0%, calculate the actual yield of 1,1,2,2-tetrabromoethane. (1 mark)
- (Relative atomic masses: H = 1.0, C = 12.0, Br = 79.9)

20. Iron can be extracted from iron pyrite (FeS_2) in two steps:



In an extraction, 12.96 g of iron pyrite was used and iron pyrite was the limiting reactant.

- (a) Calculate the theoretical yield of iron. (2 marks)
- (b) The mass of iron obtained from the extraction was 5.98 g.
- (i) Calculate the percentage yield of iron.
- (ii) Compare the actual yield with the theoretical yield and suggest TWO possible reasons for the difference. (3 marks)

(Relative atomic masses: C = 12.0, S = 32.1, Fe = 55.8)

Answers to Chapter exercise

Chapter 10

A. Fill in the blanks (p.22)

- free elements
- (a) Heating
(b) carbon
(c) Electrolysis
- extraction
- easier; earlier
- non-renewable; conserving

B. Practice questions (p.22)

- (a) zinc blende
(b) zinc oxide
(c) oxygen; sulphur dioxide; zinc oxide; zinc; carbon dioxide
- (a) Haematite
(b) coke; limestone
(c) carbon monoxide
(d) iron(III) oxide; iron(III) oxide; carbon dioxide
- (a) bauxite; aluminium oxide
(b) electrolysis; melted
(c) aluminium oxide; oxygen

C. Multiple-choice questions (p.23)

- C
Copper pyrite is a common ore of copper. Copper exists as copper(II) iron(II) sulphide (CuFeS_2) in this ore.
- B
- A
mercury(II) oxide $\rightarrow$ mercury + oxygen
- B
The more easily a metal can be extracted from its ores, the earlier it was discovered.
- D

D. Structured questions (p.23)

- (a) (Metal) ores 1
(b) In the Earth's crust 1
(c) (i) Galena 1
(ii) Lead(II) sulphide 1
- (a) Lead 1
(b) Carbon dioxide 1
(c) lead(II) oxide + carbon $\rightarrow$ lead + carbon dioxide 1
(i) Yes.
copper(II) oxide + carbon $\rightarrow$ copper + carbon dioxide 1

- (ii) No. (This is because magnesium is a reactive metal/more reactive than carbon.) 1
- (a) mercury(II) sulphide + oxygen $\rightarrow$ mercury + sulphur dioxide 1
copper(I) sulphide + oxygen $\rightarrow$ copper + sulphur dioxide 1
(b) Mercury, copper, aluminium 1
(c) Mercury, copper, aluminium 1
The more easily a metal can be extracted from its ores, the earlier it was discovered. 1

Chapter 11

A. Fill in the blanks (p.58)

- metal reactivity series
- electrons; positive; electrons
- lower
- displace
- ionic; spectator

B. Practice questions (p.58)

- (a) $4\text{Al(s)} + 3\text{O}_2\text{(g)} \rightarrow 2\text{Al}_2\text{O}_3\text{(s)}$
(b) $2\text{Al(s)} + 3\text{F}_2\text{(g)} \rightarrow 2\text{AlF}_3\text{(s)}$
(c) $2\text{C}_2\text{H}_6\text{(g)} + 7\text{O}_2\text{(g)} \rightarrow 4\text{CO}_2\text{(g)} + 6\text{H}_2\text{O(l)}$
(d) $2\text{Cu(NO}_3)_2\text{(s)} \rightarrow 2\text{CuO(s)} + 4\text{NO}_2\text{(g)} + \text{O}_2\text{(g)}$
(e) $3\text{CuO(s)} + 2\text{NH}_3\text{(g)} \rightarrow 3\text{Cu(s)} + \text{N}_2\text{(g)} + 3\text{H}_2\text{O(l)}$
(f) $4\text{FeS}_2\text{(s)} + 11\text{O}_2\text{(g)} \rightarrow 2\text{Fe}_2\text{O}_3\text{(s)} + 8\text{SO}_2\text{(g)}$
- (a) Calcium burns quite vigorously with a brick-red flame to produce a white powder.
 $2\text{Ca(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{CaO(s)}$
(b) Sodium melts to form a silvery ball and moves about quickly on the water surface. In addition, it burns with a golden yellow flame.
 $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$
(c) An intense white light is produced. A white solid forms.
 $\text{Mg(s)} + \text{H}_2\text{O(g)} \rightarrow \text{MgO(s)} + \text{H}_2\text{(g)}$
(d) Zinc dissolves and colourless gas bubbles are given out.
 $\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
(e) Magnesium dissolves and colourless gas bubbles are given out.
 $\text{Mg(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{MgSO}_4\text{(aq)} + \text{H}_2\text{(g)}$
- (a) $\text{H}^+\text{(aq)} + \text{OH}^-\text{(aq)} \rightarrow \text{H}_2\text{O(l)}$
(b) $\text{Cl}^-\text{(aq)} + \text{Ag}^+\text{(aq)} \rightarrow \text{AgCl(s)}$
(c) $\text{Mg(OH)}_2\text{(s)} + 2\text{H}^+\text{(aq)} \rightarrow \text{Mg}^{2+}\text{(aq)} + 2\text{H}_2\text{O(l)}$
(d) $\text{Pb}^{2+}\text{(aq)} + 2\text{I}^-\text{(aq)} \rightarrow \text{PbI}_2\text{(s)}$
(e) $2\text{H}^+\text{(aq)} + \text{CaCO}_3\text{(s)} \rightarrow \text{Ca}^{2+}\text{(aq)} + \text{CO}_2\text{(g)} + \text{H}_2\text{O(l)}$

C. Multiple-choice questions (p.59)

9. B

Sodium reacts vigorously with water while there is no reaction between silver and water.

10. D

11. A

12. A

13. D

14. A

15. B

P is the least reactive as it exists as a free element in nature. *Q* is more reactive than *R* as *Q* can displace *R* from the aqueous solution of a compound of *R*.

16. B

The ionic equation of the reaction is $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \longrightarrow \text{AgCl}(\text{s})$

D. Structured question (p.60)17. (a) *P*: aluminium oxide; *Q*: hydrogen;*R*: aluminium chloride;*S*: aluminium sulphate

4

(b) $2\text{Al}(\text{s}) + 3\text{H}_2\text{O}(\text{g}) \longrightarrow \text{Al}_2\text{O}_3(\text{s}) + 3\text{H}_2(\text{g})$

1

(c) $2\text{Al}(\text{s}) + 6\text{H}^+(\text{aq}) \longrightarrow 2\text{Al}^{3+}(\text{aq}) + 3\text{H}_2(\text{g})$

1

(d) Aluminium atoms lose electrons to form positive ions less readily than magnesium atoms.

1

Chapter 12**A. Fill in the blanks (p.99)**

- 6.02×10^{23} ; Avogadro constant
- molar mass
- relative atomic mass; number
- Limiting reactant
- actual yield; theoretical yield

B. Practice questions (p.99)

- $2 \times 6.02 \times 10^{23} = 1.204 \times 10^{24}$
 - $1.5 \times 2 \times 6.02 \times 10^{23} = 1.806 \times 10^{24}$
 - $\frac{2.0}{16.0 \times 2} \times 2 \times 6.02 \times 10^{23} = 7.525 \times 10^{22}$
- $2 \times 2 \times 6.02 \times 10^{23} = 2.408 \times 10^{24}$
 - $0.5 \times 3 \times 6.02 \times 10^{23} = 9.03 \times 10^{23}$
 - $\frac{24.6}{40.1 + (14.0 + 16.0 \times 3)} \times 3 \times 6.02 \times 10^{23} = 2.71 \times 10^{23}$
- $\frac{12.0}{12.0 + 1.0 \times 4} \times 100\% = 75\%$
 - $\frac{32.1}{23.0 \times 2 + 32.1 + 16.0 \times 4} \times 100\% = 22.6\%$

$$(c) \frac{16.0 \times 11}{55.8 + 32.1 + 16.0 \times 4 + 7.0 \times (1.0 \times 2 + 16.0)} \times 100\% = 63.3\%$$

$$(d) \frac{(1.0 \times 2 + 16.0) \times 10}{23.0 \times 2 + 12.0 + 16.0 \times 3 + (1.0 \times 2 + 16.0) \times 10} \times 100\% = 62.9\%$$

9. (a) Percentage by mass of H in CH_4

$$= \frac{1.0 \times 4}{12.0 + 1.0 \times 4} \times 100\% = 25\%$$

Mass of H in 10.0 g of CH_4

$$= 10.0 \text{ g} \times 25\% = 2.5 \text{ g}$$

(b) Percentage by mass of Mg in $(\text{CH}_3\text{COO})_2\text{Mg}$

$$= \frac{24.3}{(12.0 \times 2 + 1.0 \times 3 + 16.0 \times 2) \times 2 + 24.3} \times 100\% = 17.1\%$$

Mass of Mg in 28.5 g of $(\text{CH}_3\text{COO})_2\text{Mg}$

$$= 28.5 \text{ g} \times 17.1\% = 4.87 \text{ g}$$

(c) Formula mass of $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$

$$= 55.8 + 35.5 \times 3 + (1.0 \times 2 + 16.0) \times 6 = 270.3$$

Percentage by mass of Cl in $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$

$$= \frac{35.5 \times 3}{270.3} \times 100\% = 39.4\%$$

Mass of 2 mol of $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$

$$= 2 \times 270.3 \text{ g} = 540.6 \text{ g}$$

Mass of Cl in 2 mol of $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$

$$= 540.6 \text{ g} \times 39.4\% = 213 \text{ g}$$

(d) Formula mass of $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$

$$= 40.1 + 35.5 \times 2 + (1.0 \times 2 + 16.0) \times 6 = 219.1$$

Percentage by mass of H_2O in $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$

$$= \frac{(1.0 \times 2 + 16.0) \times 6}{219.1} \times 100\% = 49.3\%$$

Mass of 1.25 mol of $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$

$$= 1.25 \times 219.1 \text{ g} = 273.9 \text{ g}$$

Mass of H_2O in 1.25 mol of $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$

$$= 273.9 \text{ g} \times 49.3\% = 135.0 \text{ g}$$

10. (a)

	C	H
Mass / g	75	25
Number of moles of atoms / mol	$\frac{75}{12.0} = 6.25$	$\frac{25}{1.0} = 25$
Simplest whole number mole ratio of atoms	$\frac{6.25}{6.25} = 1$	$\frac{25}{6.25} = 4$

$\therefore$ the empirical formula of the compound is CH_4 .

(b)

	Pb	O
Mass / g	86.6	13.4
Number of moles of atoms / mol	$\frac{86.6}{207.2} = 0.418$	$\frac{13.4}{16.0} = 0.838$
Simplest whole number mole ratio of atoms	$\frac{0.418}{0.418} = 1$	$\frac{0.838}{0.418} = 2$

∴ the empirical formula of the compound is PbO_2 .

(c)

	Na	S	O
Mass / g	36.5	25.4	38.1
Number of moles of atoms / mol	$\frac{36.5}{23.0} = 1.59$	$\frac{25.4}{32.1} = 0.791$	$\frac{38.1}{16.0} = 2.38$
Simplest whole number mole ratio of atoms	$\frac{1.59}{0.791} = 2.01 \approx 2$	$\frac{0.791}{0.791} = 1$	$\frac{2.38}{0.791} = 3.01 \approx 3$

∴ the empirical formula of the compound is Na_2SO_3 .

(d)

	C	N	H	O
Mass / g	40.67	23.73	8.47	27.13
Number of moles of atoms / mol	$\frac{40.67}{12.0} = 3.39$	$\frac{23.73}{14.0} = 1.70$	$\frac{8.47}{1.0} = 8.47$	$\frac{27.13}{16.0} = 1.70$
Simplest whole number mole ratio of atoms	$\frac{3.39}{1.70} = 1.99 \approx 2$	$\frac{1.70}{1.70} = 1$	$\frac{8.47}{1.70} = 4.98 \approx 5$	$\frac{1.70}{1.70} = 1$

∴ the empirical formula of the compound is $\text{C}_2\text{NH}_5\text{O}$.

(e)

	Cu	Cl	H ₂ O
Mass / g	37.11	41.68	$100 - 37.11 - 41.68 = 21.21$
Number of moles of formula units / mol	$\frac{37.11}{63.5} = 0.584$	$\frac{41.68}{35.5} = 1.17$	$\frac{21.21}{18.0} = 1.18$
Simplest whole number mole ratio of formula units	$\frac{0.584}{0.584} = 1$	$\frac{1.17}{0.584} = 2$	$\frac{1.18}{0.584} = 2.02 \approx 2$

∴ the empirical formula of the compound is $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$.

C. Multiple-choice questions (p.100)

11. B

For (A): the substance contains 2 mol of Ag^+ ions and 1 mol of S^{2-} ions

For (B): the substance contains 1 mol of Cu^{2+} ions and 1 mol of $\text{Cr}_2\text{O}_7^{2-}$ ions

For (C): the substance contains 2 mol of Al^{3+} ions and 3 mol of SO_4^{2-} ions

For (D): the substance contains 1 mol of Mg^{2+} ions and 2 mol of NO_3^- ions

12. D

The mass of 1 mole of O_2 is 32.0 g and that of 1 mole of N_2 is 28.0 g.

13. A

14. D

Molar mass of nandrolone

$$= (12.0 \times 18 + 1.0 \times 26 + 16.0 \times 2) \text{ g mol}^{-1} \\ = 274.0 \text{ g mol}^{-1}$$

15. B

Mole ratio of X atoms to O atoms

$$= \frac{83.0}{39.1} : \frac{100 - 83.0}{16.0} \\ = 2.123 : 1.063 \\ = 2 : 1$$

16. C

$$\text{Mole ratio of ZnSO}_4 \text{ to H}_2\text{O} = \frac{2.47}{161.5} : \frac{4.40 - 2.47}{18.0} \\ = 0.0153 : 0.107 \\ = 1 : 6.99 \\ \approx 1 : 7$$

17. B

Number of moles of CaCO_3

$$= \frac{10.01}{40.1 + 12.0 + 16.0 \times 3} \text{ mol} = 0.100 \text{ mol}$$

$$\therefore \text{theoretical mass of CaO produced} = 0.100 \times (40.1 + 16.0) \text{ g} = 5.61 \text{ g}$$

18. C

$$\text{Percentage yield of the reaction} = \frac{4.31 \text{ g}}{5.61 \text{ g}} \times 100\% = 76.8\%$$

D. Structured questions (p.101)

19. (a) False

32.0 g of oxygen (O_2) contains 6.02×10^{23} molecules while 32.0 g of ozone (O_3)

$$\text{contains } \frac{32.0 \text{ g}}{16.0 \times 3 \text{ g mol}^{-1}} \times 6.02 \times 10^{23} \text{ mol}^{-1} \\ = 4.01 \times 10^{23} \text{ molecules.} \quad 1$$

(b) False

1 mol of $\text{Pb}(\text{NO}_3)_2$ contains 1 mol of Pb^{2+} ions and 2 mol of NO_3^- ions. Hence, 1 mol of $\text{Pb}(\text{NO}_3)_2$ contains 3 mol of ions. 1

20. (a) SnF_2 1
 (b) $118.7 + 19.0 \times 2 = 156.7$ 1
 (c) $\frac{19.0 \times 2}{156.7} \times 100\% = 24.3\%$ 1
 (d) $1.50 \text{ g} \times 24.3\% = 0.365 \text{ g}$ 1

Chapter 13

A. Fill in the blanks (p.135)

1. Corrosion; oxygen; water
 2. Rusting
 3. potassium hexacyanoferrate(III); phenolphthalein
 4. (a) acidic substances
 (b) soluble ionic
 (c) High
 (d) less reactive
 (e) scratched; bent; sharp
 5. (b) Electroplating
 (c) Sacrificial
 (d) Cathodic
 6. aluminium oxide; anodization

B. Practice question (p.135)

7. (a) ✗ (b) ✗
 (c) ✗ (d) ✗
 (e) ✗ (f) ✓
 (g) ✗ (h) ✓
 (i) ✗ (j) ✓

C. Multiple-choice questions (p.136)

8. D
 When copper corrodes, green solid is found on its surface. When silver corrodes, it becomes tarnished.
 9. B
 Sugar is water soluble but no ions would form when it dissolves in water. This is because sugar is not an electrolyte.
 10. C
 When an iron nail is wrapped with, attached or connected to a less reactive metal, rusting becomes faster.
 11. C
 In option (C), the iron nail is prevented from rusting by sacrificial protection.
 12. B
 13. C
 The aluminium object to be anodized is connected to the positive terminal of the d.c. power source. Dilute H_2SO_4 is used as an electrolyte.
 14. D

Distilled water contains dissolved air and the dissolved air (oxygen) can be removed from the water by boiling.

D. Structured questions (p.137)

15. (a) $4\text{Fe(s)} + 3\text{O}_2\text{(g)} + 2n\text{H}_2\text{O(l)} \rightarrow 2\text{Fe}_2\text{O}_3 \cdot n\text{H}_2\text{O(s)}$ 1
 (b) It removes water (moisture) from the air. 1
 (c) It prevents air from dissolving in the boiled distilled water. 1
 (d) (i) Tube 1 and Tube 4 1
 This is because there is no water in Tube 1 and there is no air (oxygen) in Tube 4. 1
 (ii) Tube 2
 This is because tap water contains soluble ionic compounds. The presence of dissolved ions greatly increases the conductivity of tap water. This speeds up the rusting of iron. 1
 16. (a) Galvanized iron refers to an iron that has a zinc layer coated on its surface. 1
 (b) The zinc in galvanized iron prevents iron from contacting oxygen and water. 1
 (c) No. Iron can still be protected from corrosion even the zinc layer of galvanized iron fences is damaged. This is because zinc is more reactive than iron, it will corrode (or will be oxidized) in preference to iron. 1
 (d) By painting 1
 17. (a) When aluminium is exposed to air, a thin layer of aluminium oxide quickly forms on its surface. 1
 The oxide layer sticks to the surface of aluminium very strongly and is impermeable to both oxygen and water. Hence, it can protect the aluminium from further corrosion. 1
 (b) (i) Positive terminal 1
 (ii) Negative terminal 1
 (c) The mobile phone cases can be dyed to give attractive colours. 1
 The mobile phone cases are harder. 1

Answers to Exam practice

Chapter 10

A. Multiple-choice questions (p.25)

1. A

2. D
3. D (78%)

Upon strong heating, silver oxide decomposes to silver and oxygen. The oxygen escapes from the test tube and hence the mass of the contents in the test tube decreases. As the heating proceeds, more silver oxide would have decomposed. When all the silver oxide is reduced, the mass of the contents in the test tube would remain unchanged.

B. Structured questions (p.25)

4. (a) Gold, silver, iron, aluminium 1
(b) The less reactive a metal, the more easily the metal can be extracted from its ores. 1
(c) Gold, silver, iron, aluminium 1
(d) Gold, silver, iron, aluminium 1
(e) This is because aluminium is extracted from its molten ores by electrolysis. 1
5. (a) This is because platinum is the least reactive among the four metals and exists as a free element in nature. 1
It can be extracted easily from its ores by physical separation. 1
(b) Platinum is very rare in the Earth's crust. 1
(c) Copper ores are running out/very hard to find. 1
The demand for copper is very high. 1
(d) (i) Aluminium 1
This is because it is corrosion resistant and it is relatively cheaper than copper. 1
(ii) Iron 1
This is because it is very strong and relatively cheap. 1
(iii) Platinum 1
This is because it is rare and very corrosion resistant. 1

Chapter 11

A. Multiple-choice questions (p.61)

1. B
2. C
The balanced equation for the reaction in (2) is:
(2): $2\text{MgCO}_3 + 4\text{HCl} \longrightarrow 2\text{MgCl}_2 + 2\text{CO}_2 + 2\text{H}_2\text{O}$
3. B
A more reactive metal will displace the less reactive metal from the oxide of the less reactive metal.
4. A
X is more reactive than Y because Y cannot displace X from the oxide of X upon heating.

5. B
6. B

	Reactant side	Product side	
Number of Ti atoms	1	y	(1)
Number of O atoms	2	z	(2)
Number of Cl atoms	2w	4y	(3)
Number of C atoms	x	z	(4)

From (1), $y = 1$

From (2), $z = 2$

From (4), $x = z$; $\therefore x = 2$

From (3), $2w = 4y$; $\therefore w = 2$

7. B (83%)

	Reactant side	Product side	
Number of Ni atoms	3	3	
Number of O atoms	6	6	
Number of H atoms	$6 + x$	12	(1)
Number of Cl atoms	$x + 4y$	$12 + z$	(2)
Number of Au atoms	y	y	
Net charge	-y	-z	(3)

From (1), $6 + x = 12$; $\therefore x = 6$

From (3), $-y = -z$

From (2), $x + 4y = 12 + z$; $\therefore y = 2$; $z = 2$

8. D (54%)

The more reactive a metal, the more stable is its carbonate and the carbonate is more stable to heat. Z may be zinc as the thermal decomposition of zinc carbonate gives zinc oxide, which is yellow:
 $\text{ZnCO}_3(\text{s}) \longrightarrow \text{ZnO}(\text{s}) + \text{CO}_2(\text{g})$

9. D (65%)

B. Structured questions (p.62)

10. (a) Magnesium oxide 1
(b) Magnesium burns with a very bright white light. 1
(c) $2\text{Mg}(\text{s}) + \text{O}_2(\text{g}) \longrightarrow 2\text{MgO}(\text{s})$ 1
(d) When calcium burns with oxygen, a brick-red light instead of a very bright white light is produced. The brightness of the light produced by calcium is lower.
Hence, calcium cannot be used to replace magnesium in a flashbulb. 1
11. (a) Hydrogen/ $\text{H}_2(\text{g})$ 1
(b) $\text{Ca}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \longrightarrow \text{Ca}(\text{OH})_2(\text{aq}) + \text{H}_2(\text{g})$ 1
(c) The solution turns blue because the reaction produces calcium hydroxide, which is alkaline. 2
(d) No observable change 1
Copper does not react with cold water. 1

- (e) No. Potassium reacts vigorously with water and it burns/the reaction evolves a gas very rapidly in the set-up. This poses a danger. 1
12. (a) Oxygen 1
 (b) Silver 1
 (c) Hydrogen 1
 (d) Potassium/sodium 1
 (e) (i) $\text{Zn(s)} + 2\text{HCl(aq)} \longrightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$ 1
 Or, $\text{Zn(s)} + 2\text{H}^+\text{(aq)} \longrightarrow \text{Zn}^{2+}\text{(aq)} + \text{H}_2\text{(g)}$ (1)
 (ii) $\text{Zn(s)} + \text{Cu}^{2+}\text{(aq)} \longrightarrow \text{Zn}^{2+}\text{(aq)} + \text{Cu(s)}$ 1
 (f) A, C, B 1
13. (a) Lead 1
 (b) $2\text{Al(s)} + 3\text{PbO(s)} \longrightarrow \text{Al}_2\text{O}_3\text{(s)} + 3\text{Pb(s)}$ 1
 (c) (i) Yes. $2\text{Al(s)} + \text{Fe}_2\text{O}_3\text{(s)} \longrightarrow \text{Al}_2\text{O}_3\text{(s)} + 2\text{Fe(s)}$ 1
 (ii) No. This is because aluminium is less reactive than magnesium. 1
14. (a) Zinc is more reactive than copper. 1
 $\text{Zn(s)} + \text{CuO(s)} \longrightarrow \text{ZnO(s)} + \text{Cu(s)}$ 1
 (b) Zinc is less reactive than aluminium. 1
15. (a) S, Q, R, P 1
 (b) (i) P 1
 (ii) Q 1
 (c) P 1
 (d) The blue aqueous solution turns colourless. 1
 Reddish brown solid/precipitate forms. 1
 $\text{Mg(s)} + \text{Cu}^{2+}\text{(aq)} \longrightarrow \text{Mg}^{2+}\text{(aq)} + \text{Cu(s)}$ 1

Chapter 12

A. Multiple-choice questions (p.102)

1. B
 Molar mass of the gas = $\frac{7}{0.25} \text{ g mol}^{-1} = 28.0 \text{ g mol}^{-1}$
 The molar mass of C_2H_4 is $(12.0 \times 2 + 1.0 \times 4) \text{ g mol}^{-1} = 28.0 \text{ g mol}^{-1}$
2. A
 Number of moles of H_2O in 36 g of pure water
 $= \frac{36}{1.0 \times 2 + 16.0} \text{ mol} = 2 \text{ mol}$
 Hence, 36 g of pure water contains $2 \times 3 \times 6.02 \times 10^{23}$ of atoms and contains $2 \times 6.02 \times 10^{23}$ of molecules.
3. C
 Number of moles of He in 2 g of He = $\frac{2}{4.0} \text{ mol} = 0.5 \text{ mol}$
 Number of moles of F_2 in 38 g of $\text{F}_2 = \frac{38}{19.0 \times 2} \text{ mol} = 1 \text{ mol}$

Number of molecules in 2 g of He = $0.5 \times L = y$, where L represents the Avogadro constant. This means that $L = \frac{y}{0.5}$.

Hence, number of molecules in 38 g of $\text{F}_2 = 1 \times \frac{y}{0.5} = 2y$.

4. D (60%)

Number of moles of SO_2 in 8.0 g of SO_2
 $= \frac{8.0}{32.0 + 16.0 \times 2} \text{ mol} = 0.125 \text{ mol}$

Number of moles of O_2 in 2.0 g of O_2
 $= \frac{2.0}{16.0 \times 2} \text{ mol} = 0.0625 \text{ mol}$

Number of molecules in 8.0 g of $\text{SO}_2 = 0.125 \times L = n$, where L represents the Avogadro constant. This means that $L = \frac{n}{0.125}$.

Hence, number of molecules in 2.0 g of $\text{O}_2 = 0.0625 \times \frac{n}{0.125} = 0.50n$.

5. C

Percentage by mass of Ca in CaCO_3

$$= \frac{40.1}{40.1 + 12.0 + 16.0 \times 3} \times 100\% = 40.0\%$$

Percentage by mass of Ca in $\text{Ca(NO}_3)_2$

$$= \frac{40.1}{40.1 + (14.0 + 16.0 \times 3) \times 2} \times 100\% = 24.4\%$$

Percentage by mass of Ca in Ca(OH)_2

$$= \frac{40.1}{40.1 + (16.0 + 1.0) \times 2} \times 100\% = 54.1\%$$

Percentage by mass of Ca in CaSO_4

$$= \frac{40.1}{40.1 + 32.1 + 16.0 \times 4} \times 100\% = 29.4\%$$

6. C

Percentage by mass of Fe in Fe_2O_3

$$= \frac{55.8 \times 2}{55.8 \times 2 + 16.0 \times 3} \times 100\% = 69.9\%$$

Mass of Fe_2O_3 present in 100 g of iron ore = $100 \times 70\% = 70 \text{ g}$

$\therefore$ mass of Fe that can be extracted = $70 \text{ g} \times 69.9\% = 48.9\%$

7. C (82%)

Mass of O in 24.0 g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

$$= 24.0 \text{ g} \times \frac{16.0 \times 9}{63.5 + 32.1 + 16.0 \times 4 + 5 \times (1.0 \times 2 + 16.0)} = 13.8 \text{ g}$$

8. D

$$\begin{aligned} \text{Mole ratio of X to H}_2\text{O} &= \frac{100 - 62.9}{106.0} : \frac{62.9}{18.0} \\ &= 0.35 : 3.49 \\ &= 1 : 9.97 \\ &\approx 1 : 10 \end{aligned}$$

9. C (55%)

Let the relative atomic mass of X be a .

$$\frac{a}{39.1 \times 2 + a + 16.0 \times 4} = \frac{26.8}{100}$$

$$100a = (142.2 + a) \times 26.8$$

$$a = 52.0$$

From the Periodic Table, X is Cr. It is a transition metal and is in the fourth period of the Periodic Table.

10. A

Mole ratio of X atoms to O atoms

$$= \frac{8.42 - 2.40}{40.1} : \frac{2.40}{16.0}$$

$$= 0.150 : 0.150$$

$$= 1 : 1$$

11. A

From the equation, mole ratio of SiCl_4 to Si = 1 : 1.

$$\text{Theoretical yield of Si} = \frac{8.50}{28.1 + 35.5 \times 4} \times 28.1 \text{ g} = 1.40 \text{ g}$$

$$\text{Mass of Si made} = 1.40 \text{ g} \times 90\% = 1.26 \text{ g}$$

12. A

Let the relative atomic mass of X be a .

$$\text{Number of moles of X} = \frac{0.80}{a} \text{ mol}$$

$$\text{Number of moles of O} = \frac{0.40}{16.0} \text{ mol} = 0.025 \text{ mol}$$

$$\text{Mole ratio of X : O} = \frac{\frac{0.80}{a}}{\frac{0.40}{16.0}} = \frac{2}{3}$$

$$a = 48.0$$

$\therefore$ X is titanium. (The relative atomic mass of titanium is 47.9)

13. D (62%)

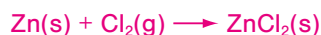
N mol of Fe_2O_3 requires $3N$ mol of CO for complete reaction. However, only $2N$ mol of CO is available. Hence, CO is the limiting reactant.

From the equation, mole ratio of CO to Fe = 3 : 2.

$$\therefore \text{number of moles of Fe formed} = \frac{2N}{3} \times 2 = \frac{4N}{3}$$

14. C

Zinc reacts with chlorine according to the following equation:



For (A): 1 mol of Zn atoms requires 1 mol of chlorine molecules for complete reaction.

$$\text{For (B): 1 g of Zn requires } \frac{1 \text{ g}}{65.4 \text{ g mol}^{-1}} \times (35.5 \times 2) \text{ g mol}^{-1} = 1.09 \text{ g of chlorine for complete reaction.}$$

$$\text{For (C): percentage by mass of Cl in ZnCl}_2 = \frac{35.5 \times 2}{65.4 + 35.5 \times 2} \times 100\% = 52.1\%$$

$$\text{For (D): molar mass of ZnCl}_2 = 65.4 + 35.5 \times 2 \text{ g mol}^{-1} = 136.4 \text{ g mol}^{-1}$$

15. B

Number of moles of O atoms in the oxide

$$= \frac{2.4}{16.0} \text{ mol} = 0.15 \text{ mol}$$

Number of moles of Y atoms = 0.1 mol (given)

Mole ratio of Y atoms to O atoms = 0.1 : 0.15 = 2 : 3

So, the formula of the oxide is Y_2O_3 . Y might be aluminium because the ratio of Al atoms to O atoms in the empirical formula of aluminium oxide is 2 : 3.

B. Structured questions (p.104)

16.

	C	H	O
Mass / g	63.1	5.3	31.6
Number of moles of atoms / mol	$\frac{63.1}{12.0} = 5.26$	$\frac{5.3}{1.0} = 5.30$	$\frac{31.6}{16.0} = 1.98$
Simplest whole number mole ratio of atoms	$\frac{5.26}{1.98} = 2.66$	$\frac{5.30}{1.98} = 2.68$	$\frac{1.98}{1.98} = 1$
	8	8	3

$\therefore$ the empirical formula of vanillin is $\text{C}_8\text{H}_8\text{O}_3$. 1

Let the molecule formula of vanillin be $(\text{C}_8\text{H}_8\text{O}_3)_n$, where n is an integer.

$$n \times (12.0 \times 8 + 1.0 \times 8 + 16.0 \times 3) = 152.0$$

$$n = 1$$

$\therefore$ the molecular formula of vanillin is $\text{C}_8\text{H}_8\text{O}_3$. 1

17.

	C	N	H	O
Mass / g	63.58	9.27	5.96	21.19
Number of moles of atoms / mol	$\frac{63.58}{12.0} = 5.30$	$\frac{9.27}{14.0} = 0.662$	$\frac{5.96}{1.0} = 5.96$	$\frac{21.19}{16.0} = 1.32$
Simplest whole number mole ratio of atoms	$\frac{5.30}{0.662} = 8$	$\frac{0.662}{0.662} = 1$	$\frac{5.96}{0.662} = 9$	$\frac{1.32}{0.662} = 2$

$\therefore$ the empirical formula of paracetamol is



Let the molecule formula of paracetamol be $(\text{C}_8\text{NH}_9\text{O}_2)_n$, where n is an integer.

$$n \times (12.0 \times 8 + 14.0 + 1.0 \times 9 + 16.0 \times 2) = 151.0$$

$$n = 1$$

$\therefore$ the molecular formula of paracetamol is



18. (a) $\text{Mg(s)} + \text{H}_2\text{O(g)} \longrightarrow \text{MgO(s)} + \text{H}_2(\text{g})$ 1

- (b) From the equation, mole ratio of Mg to MgO = 1 : 1.
Theoretical mass of MgO produced

$$= \frac{4.86}{24.3} \times (24.3 + 16.0) \text{ g}$$

$$= 8.06 \text{ g}$$
1
1
19. (a) Number of moles of C_2H_2

$$= \frac{2.00}{12.0 \times 2 + 1.0 \times 2} \text{ mol} = 0.0769 \text{ mol}$$
Number of moles of Br_2

$$= \frac{5.20}{79.9 \times 2} \text{ mol} = 0.0325 \text{ mol}$$
0.0325 mol of Br_2 requires only $\frac{0.0325}{2} \text{ mol}$

$$= 0.0163 \text{ mol of } \text{C}_2\text{H}_2 \text{ for complete reaction.}$$
Hence, Br_2 is the limiting reactant.
From the equation, mole ratio of Br_2 to $\text{C}_2\text{H}_2\text{Br}_4$ = 2 : 1.
Theoretical yield of $\text{C}_2\text{H}_2\text{Br}_4$

$$= 0.0325 \times \frac{1}{2} \times (12.0 \times 2 + 1.0 \times 2 + 79.9 \times 4) \text{ g}$$

$$= 5.62 \text{ g}$$
1
1
- (b) Actual yield of $\text{C}_2\text{H}_2\text{Br}_4$ = 5.62 g $\times$ 90%

$$= 5.06 \text{ g}$$
1
20. (a) From the equations, 1 mol of FeS_2 produces 1 mol of Fe.
Theoretical yield of Fe

$$= \frac{12.96}{55.8 + 32.1 \times 2} \times 55.8 \text{ g}$$

$$= 6.03 \text{ g}$$
1
1
- (b) (i) Percentage yield of Fe

$$= \frac{5.98 \text{ g}}{6.03 \text{ g}} \times 100\% = 99.2\%$$
1
- (ii) Any TWO of the following:
 - The reactions may not complete. 1
 - Impurities are present in the iron pyrite. 1
 - Some iron is lost during transfer or weighing. 1
(Accept other reasonable answers.) 1

Chapter 13

A. Multiple-choice questions (p.139)

- C
Rusting refers to the corrosion of iron only.
- B (87%)
The iron hook in contact with a less reactive metal (e.g. copper) would corrode faster.
- D
Zinc offers sacrificial protection to iron for preventing the crash barrier from rusting.
- B (86%)

- A
In cathodic protection, a conductor such as graphite or platinum alloy is connected to the positive terminal of the d.c. power supply.
- D (93%)
- D (55%)
In option (D), the iron nail corrodes fastest. This is because the iron nail is connected to a less reactive metal and is immersed in an aqueous solution of a soluble ionic compound. Besides, the positive pole of the d.c. power supply draws electrons away from the iron nail in the set-up. This also promotes the formation of $\text{Fe}^{2+}(\text{aq})$.
- A (66%)
The oxide of aluminium and that of iron are both insoluble in water.

B. Structured questions (p.140)

- (a) $\text{Fe}_2\text{O}_3 \cdot n\text{H}_2\text{O}$ 1
(b) The rust formed is just loosely attached to the iron surface. It falls off from the iron surface easily. 1
When the fresh iron surface is exposed to air, it reacts with oxygen and water. Thus, rusting continues until the iron object corrodes completely. 1
- (i) Tin layer prevents iron from contacting oxygen and water. 1
(ii) Magnesium is more reactive than iron. It will lose electrons (corrode) instead of iron. 1
OR
Magnesium blocks offer sacrificial protection to iron for preventing underground water pipelines from rusting. (1)
(iii) The car battery provides electrons to the car body, preventing iron from forming iron(II) ions. 1
- (a) Tube 1: iron nail rusts because air (oxygen) and water are present. 1
Tube 2: iron nail does not rust because there is no water. 1
Tube 3: iron nail does not rust because there is no air (oxygen). 1
Tube 4: iron nail does not rust because the paint can serve as a protective layer that prevents the iron nail from contacting oxygen and water. 1
Tube 5: iron nail does not rust because stainless steel is resistant to corrosion. 1

- (b) Magnesium is more reactive than iron. 1
It provides sacrificial protection to iron for preventing steel from rusting. 1
- (c) Aluminium has a coating of aluminium oxide. 1
It protects the metal from further corrosion. 1
(The author is responsible for the solutions and that (a) they have neither been provided nor approved by AQA and (b) they may not necessarily constitute the only possible solutions.)
11. (a) Steel is mainly made up of iron. The iron reacts with oxygen and water as the cables are exposed to air. 1
- (b) Steel is much stronger than aluminium. It can support a heavier load (weight). 1
- (c) (i) The paint would be scratched off when cable cars move. 1
(ii) The oil can serve as a protective layer that prevents the iron from contacting oxygen and water. 1
(iii) Oil can also serve as a lubricant. 1
(iv) Repeated applications are required. 1
OR
Dust would stick to the oil. (1)
12. (a) Oxygen and water 2
(b) Zinc 1
(c) Magnesium is higher than iron in the reactivity series. It would lose electrons (corrode) instead of iron. 1
Copper is lower than iron in the reactivity series. Placing copper blocks on the ship's hull would speed up the corrosion of iron. 1
- (d) (i) $\text{Fe}_2\text{O}_3 + 2\text{H}_3\text{PO}_4 \longrightarrow 2\text{FePO}_4 + 3\text{H}_2\text{O}$ 1
(ii) Iron(III) phosphate acts as a barrier which prevents iron from contacting oxygen and water. 1
- (Cambridge Assessment International Education bears no responsibility for the example answers to questions taken from its past question papers which are contained in this publication. In examinations, the way marks are awarded may be different.)

Answers to Part exercise

Chapter 13

A. Multiple-choice questions (p.142)

1. D
In general, the more easily the metal that can be extracted from its ore, the earlier it was discovered.

2. C
Potassium burns with a lilac flame and hydrogen is produced when it is added to water. The reaction is exothermic.
3. A
4. B
A more reactive metal can displace any less reactive metal from an aqueous solution of the compound of the less reactive metal.
5. A
Number of moles of atoms in 28.0 g of N_2

$$= \frac{28.0}{14.0 \times 2} \text{ mol} \times 2 = 2 \text{ mol}$$
 Number of moles of atoms in 2.0 g of H_2

$$= \frac{2.0}{1.0 \times 2} \text{ mol} \times 2 = 2 \text{ mol}$$
 Number of moles of atoms in 18.0 g of O_2

$$= \frac{18.0}{16.0 \times 2} \text{ mol} \times 2 = 1.13 \text{ mol}$$
 Number of moles of atoms in 34.0 g of Cl_2

$$= \frac{34.0}{35.5 \times 2} \text{ mol} \times 2 = 0.958 \text{ mol}$$
 Number of moles of atoms in 40.0 g of Ar

$$= \frac{40.0}{40.0} \text{ mol} = 1 \text{ mol}$$
6. D
Molar mass of the substance

$$= \frac{36 \text{ g}}{0.4 \text{ mol}} = 90.0 \text{ g mol}^{-1}$$
7. D
Mole ratio of NiSO_4 to H_2O

$$= \frac{2.29}{154.8} : \frac{3.91 - 2.29}{18.0}$$

$$= 0.0148 : 0.09$$

$$= 1 : 6.08$$

$$\approx 1 : 6$$
8. A
Let the relative atomic mass of metal M be a .
For oxide X , mole ratio of M atoms to O atoms

$$= \frac{3.760}{a} : \frac{4.707 - 3.760}{16.0}$$

$$= 1 : 1$$

$$\therefore a = 63.5$$
 For oxide Y , mole ratio of M atoms to O atoms

$$= \frac{3.760}{63.5} : \frac{4.234 - 3.760}{16.0}$$

$$= 0.0592 : 0.0296$$

$$= 2 : 1$$

$$\therefore \text{the formula of oxide } Y \text{ is } M_2\text{O}.$$
9. D
Number of moles of Al

$$= \frac{1.00}{27.0} \text{ mol} = 0.0370 \text{ mol}$$
 Number of moles of Cr_2O_3

$$= \frac{4.00}{52.0 \times 2 + 16.0 \times 3} \text{ mol} = 0.0263 \text{ mol}$$

As 0.0370 mol of Al requires only $\frac{0.0370}{2}$ mol = 0.0185 mol of Cr_2O_7 for complete reaction. Hence, Cr_2O_3 is in excess.

Theoretical yield of Cr = $0.0370 \times 52.0 \text{ g} = 1.924 \text{ g}$

$\therefore$ percentage yield of the reaction

$$= \frac{1.75 \text{ g}}{1.924 \text{ g}} \times 100\% = 91.0\%$$

10. C

Let the relative atomic mass of metal T be a .



$$\text{Number of moles of } T = \frac{4.86}{a} \text{ mol}$$

Number of moles of $T\text{Cl}_2$

$$= \frac{19.06}{a + 35.5 \times 2} \text{ mol}$$

From the equation, 1 mole of T produces 1 mole of $T\text{Cl}_2$.

$$\therefore \frac{4.86}{a} : \frac{19.06}{a + 71.0} = 1 : 1$$

$$a = 24.3$$

11. D (38%)

Oxygen is the limiting reactant as the copper or iron used is in excess here.

Helium has no reaction with the metals.

12. D

Fe can form Fe^{2+} and Fe^{3+} ions while Mg can form only Mg^{2+} ions.

B. Structured questions (p.143)

13. (a) Potassium compounds are mostly soluble in water. Thus, these compounds dissolve in the water body forming dissolved salts. 1
Calcium carbonate is water insoluble. Thus, it is found in sediments. 1
- (b) The electronic arrangements of potassium and calcium atoms are 2,8,8,1 and 2,8,8,2 respectively. 1
In order to attain the stable electronic arrangement of argon atom, a potassium atom has to lose only one outermost shell electron while a calcium atom has to lose two. 1
- (c) Both potassium and calcium are reactive metals. They are both extracted by electrolysis of the molten metal ores. However, electrolysis can only be used after the invention of electricity. 1
- (d) For potassium compounds: lilac 1
For calcium compounds: brick-red 1

14. (a) (i) The iron wool reacted with oxygen in the air. 1
As the oxygen inside the burette was consumed, the pressure inside the burette was reduced. The atmospheric pressure forced the water in the trough into the burette. 1
- (ii) Fe changed to $\text{Fe}_2\text{O}_3 \cdot n\text{H}_2\text{O}$ in rusting. 1
 $\text{Fe}_2\text{O}_3 \cdot n\text{H}_2\text{O}$ has a higher formula mass than Fe. 1
- (b) (i) To study the effect of temperature on the rate (speed) of rusting 1
(ii) Independent variable: temperature 1
Dependent variable: mass of iron wool after the experiment 1
Control variable: mass of iron wool at the start of the experiment/volume of air inside the burette at the start of the experiment 1
- (iii) The rate (speed) of rusting increases with temperatures. 1
15. (a) X, iron, Y 1
Y can displace both iron and X from their respective chloride solutions. 1
Iron can displace X from the chloride solution of X, whereas it cannot displace Y from the chloride solution of Y. 1
- (b) (i) $\text{Y(s)} + \text{FeCl}_2\text{(aq)} \longrightarrow \text{YCl}_2\text{(aq)} + \text{Fe(s)}$ 1
OR
 $\text{Y(s)} + \text{Fe}^{2+}\text{(aq)} \longrightarrow \text{Y}^{2+}\text{(aq)} + \text{Fe(s)}$ (1)
- (ii) $\text{Fe(s)} + \text{XCl}_2\text{(aq)} \longrightarrow \text{FeCl}_2\text{(aq)} + \text{X(s)}$ 1
OR
 $\text{Fe(s)} + \text{X}^{2+}\text{(aq)} \longrightarrow \text{Fe}^{2+}\text{(aq)} + \text{X(s)}$ (1)
- (c) Displacement reaction 1
- (d) Y 1
This is because Y is more reactive than iron and it would lose electrons (corrode) instead of iron. 1

Answers to Revision test

Chapter 13

A. Multiple-choice questions (p.146)

- D
- B
- B

4. D

	Reactant side	Product side	
Number of Cu atoms	x	x	
Number of H atoms	y	8	(1)
Number of N atoms	y	2x + 2	(2)
Number of O atoms	3y	6x + 6	

From (1), $y = 8$

From (2), $y = 2x + 2$

i.e. $8 = 2x + 2$

$\therefore x = 3$

5. B

6. D

1 mole of C_2H_4 contains 4L hydrogen atoms.

Let the number of moles of C_2H_4 that contains n hydrogen atoms be x .

By simple proportion, $\frac{1}{x} = \frac{4L}{n}$

$x = \frac{n}{4L}$

7. C

Mass of water of crystallization in 5.0 g of $CuSO_4 \cdot 5H_2O$

$$= 5.0 \text{ g} \times \frac{(1.0 \times 2 + 16.0) \times 5}{63.5 + 32.1 + 16.0 \times 4 + (1.0 \times 2 + 16.0) \times 5}$$

$$= 1.8 \text{ g}$$

8. C

$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$

Mass of $MgSO_4$ produced

$$= \frac{3.0}{24.3} \times (24.3 + 32.1 + 16.0 \times 4) \text{ g} = 14.9 \text{ g}$$

9. C

10. A

11. D

The reactivity of the elements in Group I of the Periodic Table increases down the group.

B. Structured questions (p.147)

12. (a) Sodium burns with a golden yellow flame while calcium does not burn. 1

Sodium moves about quickly on the water surface while calcium sinks in water. 1

(b) (i) $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$ 1

(ii) $Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(s) + H_2(g)$ 1

(c) Each sodium atom has to lose one outermost shell electron in order to attain a stable octet electronic arrangement but each calcium atom has to lose two. 1

OR

Each sodium atom has fewer protons (positively charged) in the nucleus than each calcium atom. The outermost shell electron in the sodium atom is less strongly bound to the nucleus compared to those in the calcium atom. (1)

13. (a) $2ZnS(s) + 3O_2(g) \rightarrow 2ZnO(s) + 2SO_2(g)$ 1

(b) $2ZnO(s) + C(s) \rightarrow 2Zn(s) + CO_2(g)$ 1

Carbon is used as a reducing agent in this step. 1

(c) Number of moles of ZnS
 $= \frac{2437.5}{65.4 + 32.1} \text{ mol} = 25.0 \text{ mol}$ 1

Number of moles of C
 $= \frac{480.0}{12.0} \text{ mol} = 40.0 \text{ mol}$ 1

From the equations, 25.0 mol of ZnS requires only $\frac{25.0}{2} \text{ mol} = 12.5 \text{ mol}$ of C for complete reaction. Hence, C is in excess.

Theoretical yield of Zn
 $= 25.0 \times 65.4 \text{ g} = 1635 \text{ g}$ 1

(d) (i) Sacrificial protection 1

(ii) Zinc loses electrons more readily than iron. It 'sacrifices' itself while the iron does not rust. 1

(iii) Magnesium 1

(iv) Cathodic protection 1

14. (a) Town gas is flammable. 1

	Cu	O
Mass / g	22.16 – 20.10 = 2.06	22.68 – 22.16 = 0.52
Number of moles of atoms / mol	$\frac{2.06}{63.5} = 0.0324$	$\frac{0.52}{16.0} = 0.0325$
Simplest whole number mole ratio of atoms	$\frac{0.0324}{0.0324} = 1$	$\frac{0.0325}{0.0324} = 1$

$\therefore$ the empirical formula of the oxide of copper is CuO. 3

(2 marks for correct calculations steps)

(c) $CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(g)$ 1
 OR

$CuO(s) + CO(g) \rightarrow Cu(s) + CO_2(g)$ (1)

(d) This is to prevent the hot copper formed from reacting with oxygen again. 1